



Global public health intelligence report 2023



World Health
Organization

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ISBN 978-92-4-009820-6 (electronic version)

ISBN 978-92-4-009821-3 (print version)

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Suggested citation. Global public health intelligence report 2023. Geneva: World Health Organization; 2024. Licence: CC BY-NC-SA 3.0 IGO.

Cataloguing-in-Publication (CIP) data. CIP data are available at: <https://apps.who.int/iris>.

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Cover: *top:* A WHO epidemiologist conducting public health intelligence training through the Epidemic Intelligence from Open Sources (EIOS) system signals in Osong, Republic of Korea, 2019 © WHO/Kim Yunsoo — *left:* WHO Polio Eradication officers Dr Samreen Khalil and Dr Farukh S. along with Irfanullah K. and Zaqir H. conducting environmental sampling for polio in Peshawar, Khyber Pakhtunkhwa, Pakistan, 2020 © WHO/Blink Media: Saiyna Bashir — *right:* A WHO staff member updates the information on a whiteboard as the team compiles crucial information on confirmed Ebola cases in Freetown, Western Area Urban District, Sierra Leone, 2015 © WHO/Dante Licona

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FOREWORD

Early warning and response are key to tackle emerging and acute public health risks globally. Therefore, the World Health Organization (WHO) has implemented a comprehensive method to public health intelligence (PHI) for the global detection, verification and risk assessment of acute public health threats.

Public health intelligence activities at WHO are conducted 24 hours a day at WHO headquarters and all six regional offices, throughout every day of the year, and are underpinned by the International Health Regulations (2005). These regulations mandate countries to enhance their surveillance efforts and assess, verify and report events that could potentially be a public health emergency of international concern (PHEIC).

WHO has an all-hazards approach to detecting signals of potential public health events, which means that threats of any nature – including infectious diseases, but also natural disasters, geohazards and human-made conflicts that can cause public health events of international concern – are assessed.

This report is the third global report on public health intelligence activities at WHO. It focuses on activities conducted in 2023, and builds upon the bi- and tri-regional reports previously published by the WHO Region of the Americas, the WHO European Region and the WHO African Region.

Public health threats occur daily across the world, with several thousand signals detected and verified annually by WHO's public health intelligence activities. A proportion of these signals become events, nearly 300 per year on average, and their risks are assessed. Furthermore, nearly 240 information products, such as bulletins, announcements and Disease Outbreak News reports, are published annually by WHO to provide Member States and the public with factual and timely epidemiological information on substantiated acute public health events or potential events of concern.

The report also underscores how collaboration is crucial to the public health intelligence activities carried out by WHO. In particular, the importance of collaboration and engagement between WHO and States Parties under the International Health Regulations (2005), aligning



with the framework to strengthen the global architecture for health emergency preparedness, response and resilience (HEPR), set out at the Seventy-fifth World Health Assembly in May 2022.

Under this framework, collaborative surveillance has been identified as one of the five interconnected systems enabling effective prevention, preparation for, detection, response to, and recovery from health emergencies at subnational, national, regional and global levels.

Collaborative surveillance is defined as the systematic strengthening of capacity and collaboration among diverse stakeholders, both within and beyond the health sector, with the ultimate goal of enhancing public health intelligence and improving evidence for decision-making.

In the coming years, as new, (re)-emerging, and recurring health threats continue to arise and the consequences of the climate crisis worsen, the significance of public health intelligence will become even more pronounced, with an increased need for collaboration and local connections to ensure active engagement, enhanced information sharing and rapid verification of acute public health events.

The World Health Organization will continue in leading the way and making continuous efforts to enhance public health intelligence operations, advocate for best practices, and share valuable evidence for the betterment of individuals and communities worldwide.



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ACKNOWLEDGEMENTS

The *Global public health intelligence report 2023* is the third Global public health intelligence report. It builds on the two previous *Global public health intelligence reports*¹ and previous bi- and tri-regional *Public health intelligence reports*², produced by the World Health Organization Regional Office for the Americas, the World Health Organization Regional Office for Europe and the World Health Organization Regional Office for Africa.

This report is a collective work by all Public Health Intelligence (PHI) teams in the World Health Organization regional offices and in the Department of Alert and Response Coordination, Emergency Response Division, World Health Organization Health Emergencies Programme.

A core team produced the initial draft and coordinated the response and feedback to the report. They included Neil J. Saad-Duque, Alessandro Miglietta, Adedoyin Awofisayo-Okuyelu, Krista Swanson, Lidia Alexandrova Ezerska, Maria Almiron and Esther Hamblion.

Many people provided feedback and contributed to the data collection, analyses, preparation of figures and tables, and writing of the final version of the report. We would like to acknowledge past members of World Health Organization Public Health Intelligence teams across the Organization and particularly want to acknowledge the contribution of the following World Health Organization colleagues:

WHO African Region

Etien Koua, George Sie Williams, Jean Paul Kimenyi, Ramazani Mangosa Zaza, Freddy Kambale Kavoga, Joyce Nguna, Fordah Sarah, Donkor Gianni-ferrari

WHO Region of the Americas/Pan American Health Organization

Florence Heuschen, Alison Paredes, Diana Castillo, Consuelo Pinzon, Christian Hertlein, Ezequiel Travin, Raul Garcia Acevedo, Silvano Barbosa de Oliveira, Juan Del Valle, V. Guillermo Sequera, Camilo Avila, Katia Uchimura, Ghabriela Boitrage, Rayane Barbosa, Gabriely Mota Costa, Fabianne de Castro Teixeira, Dalila Machado Botelho Oliveira

1. See: https://cdn.who.int/media/docs/default-source/documents/emergencies/phi_2021_annual_report_web.pdf?sfvrsn=7482a7c9_4&download=true and <https://www.who.int/publications/i/item/9789240073579>.

2. See: <https://www.paho.org/en/dva-annual-report>.

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Anastasia Smirnova, Blanche Greene-Cramer, Bruno Rocha Evia, Dubravka Selenic Minet, Egmond Evers, Etsub Engedashet Tachelew, Felix Moek, Kaja Kaasik-Aaslav, Ketevan Glonti, Marion Buch-Jorgensen, Tatiana Metcalf, Vaishali Sogadar, Victoria Ndarukwa, Yosef Temre

ABBREVIATIONS

COVID-19	coronavirus disease 2019
DON	Disease Outbreak News
EIS	Event Information Site for IHR National Focal Points
EIOS	Epidemic Intelligence from Open Sources
EMS	Event Management System
ERF	Emergency Response Framework
FAO	Food and Agriculture Organization of the United Nations
GLEWS+	Joint FAO–WHO–WOAH Global Early Warning System
GOARN	Global Outbreak, Alert and Response Network
IHR (2005)	International Health Regulations (2005)
MERS	Middle East respiratory syndrome
NFP	National International Health Regulations Focal Point
PHEIC	Public Health Emergency of International Concern
PHSA	Public Health Situation Analysis
RRA	Rapid Risk Assessment
WHO	World Health Organization
WOAH	World Organisation for Animal Health

GLOSSARY

DETECTION

It is a process by which events of potential international public health concern are identified and recorded. WHO detects signals (see *Signal*) using various sources. These include: (i) media through the Epidemic Intelligence from Open Sources (EIOS) system; (ii) reporting through governmental channels, such as the Ministry of Health, national or sub-national government agencies and public health centres or institutes; (iii) other United Nations agencies; (iv) partner networks, such as the Global Outbreak, Alert and Response Network (GOARN); and (v) the Joint FAO–WHO–WOAH Global Early Warning System (GLEWS+). Events are also directly reported by States Parties to WHO through National International Health Regulations Focal Point (NFP) or comparable (European Commission (EC) Early Warning and Response System (EWRS) for EU/EEA Member States) channels.

DISEASE OUTBREAK NEWS

Disease Outbreak News (DON) reports provide verified information on acute public health events to the public, which are shared by WHO under the International Health Regulations (2005) [IHR (2005)]. They provide validated epidemiological information on confirmed or potential acute public health events that have significant public health implications or the potential to become a public health emergency of international concern (PHEIC).

EVENT

Event is defined by Article 1 of the IHR (2005) as “a manifestation of disease or an occurrence that creates a potential for disease”. This can include events that are infectious, zoonotic, food safety, chemical, radiological or nuclear in origin and transmitted by persons, vectors, animals, goods/food or through the environment. Due to inherent differences in pathogen presence, epidemiological dynamics, socio-political factors, climatic conditions and other disease drivers, what might constitute an event in one geographical area at a certain time might not be sufficient to meet the threshold for an event in another area, at a comparable or different time. Therefore, the categorization of events will vary between WHO regions. When comparing events between WHO regions, the underlying dynamics that influence the emergence, spread and impact of acute public health events should always be taken into account.

EVENT DESIGNATION

Event designation is assigned during the risk assessment process to events recorded in the internal WHO Event Management System (EMS) (see “Event Management System”). There are six designation categories:

1. *Substantiated*, when the presence of a hazard is confirmed, or the number of human cases exceeds established thresholds.

2. *Discarded*, when no international risk is expected or, after assessment, not deemed to be a significant public health risk.
3. *No outbreak*, when no human cases are reported (e.g., in instances of outbreaks among animals), or when the number of human cases reported does not meet the threshold of an outbreak, or when the occurrence of the hazard is within expected levels.
4. *Unverifiable*, when no information is forthcoming from the NFP or responsible national authority to substantiate its occurrence, despite the best efforts to obtain such information.
5. *Under verification*, when the event continues to be undergoing the process of verification.
6. *No designation*, when no information regarding the designation is available.

EVENT MANAGEMENT SYSTEM

Event Management System (EMS) is the central electronic repository for event-related information. The EMS is a password-protected web-based tool accessible to designated personnel at WHO country and regional offices, and headquarters, subject to completion of relevant training. The key objectives of EMS are to support WHO users in the sharing and analysing of information on events as well as reviewing the assessed risk to guide public health actions. There is a linkage between the EMS and EIS to facilitate information sharing. The EMS is not intended to be an exhaustive database of all acute public health events occurring worldwide. Rather, its objective is to support the internal management of WHO for acute public health events.

HAZARD

Hazard³ refers to the nature or type of the event. WHO has adopted an all-hazards approach, in which events of any nature, including infectious diseases but also disasters, chemical, biological or nuclear threats, and human-made conflicts, are assessed. Hazards have been grouped into seven categories, which include the following:

1. *Animal or zoonosis*, if there is potential harm to public health from an animal or zoonosis.
2. *Chemical*, if there is potential harm to public health from the toxic effects of chemical substances, which are chiefly non-medical, as to source.
3. *Disaster*, if there is potential harm to public health from a natural disaster.
4. *Food safety*, if there is potential harm to public health from the toxic effects of food (poisoning or injury).
5. *Infectious*, if there is potential harm to public health from an infectious disease.
6. *Product*, if there is potential harm to public health from contaminated or faulty therapeutic goods including medicines, blood products, tissues and organs, medical devices, diagnostic tests and devices, etc. — including poisonings due to mislabelling of therapeutic goods.

3. During 2022, select hazard types were reclassified in the EMS. The following hazard classifications that were used in previous reports (animal, zoonosis, and food safety) were reclassified to other respective hazards, and therefore, are no longer shown as separate classifications.

7. *Other*, which includes the following hazards:
 - a. *Nutritional deficiency*, if there is potential harm to public health from nutritional deficiencies.
 - b. *Radionuclear*, if there is potential harm to public health from the toxic effects of ionizing radiation.
 - c. *Societal*, if there are violent or hostile events, caused by humans, which have an impact on public health.
 - d. *Noncommunicable*, if there is potential harm to public health from a noncommunicable disease.
 - e. *Undetermined*, if there is potential harm to public health from an undetermined hazard.

NATIONAL IHR FOCAL POINT

National IHR Focal Point (NFP) is defined by Article 1 of the IHR (2005) as “the national centre, designated by each State Party, which shall be accessible at all times for communications with WHO IHR Contact Points under these Regulations”. These are established as centres or offices rather than individuals.

PUBLIC HEALTH INTELLIGENCE

Public health intelligence (PHI) is defined by the United Nations terminology database as “a core public health function responsible for identifying, collecting, connecting, synthesizing, analysing, assessing, interpreting and generating a wide range of information for actionable insights and disseminating these for informed and effective decision-making to protect and improve the health of the population”.

PUBLIC HEALTH SITUATION ANALYSES

The specific purpose of the public health situation analyses (PHSA) is to identify the current health status and potential health threats that the population may face, the functioning of the health system and the humanitarian health response, as defined in the *WHO Emergency Response Framework* (ERF 2.1). It is a review of the latest available secondary data. The PHSA aims to provide all health sector partners, including local and national authorities, nongovernmental organizations (NGOs), donor agencies and United Nations agencies with a common and comprehensive understanding of the public health situation in a crisis in order to inform evidence based collective humanitarian health response planning. The PHSA may also be used to feed other sectoral and intersectoral products, such as providing the health input to the Humanitarian Needs Overview, and is also used in support of the WHO (re-)grading process.

RISK ASSESSMENT

Risk assessment is conducted at various stages of public health intelligence activities, including at the signal detection stage, when a signal is designated as an event, or an event is notified to WHO. For all events the risk assessment is documented at the stage when the event is first entered into EMS against the following criteria: the potential public health impact, risk of international spread, risk of trade or travel restrictions, and if the event is considered epidemiologically unusual. This risk assessment is repeated whenever new information on the event becomes available.

RAPID RISK ASSESSMENT

Rapid Risk Assessment (RRA) is a formalized in-depth internal WHO risk assessment. It assesses the likelihood and consequences of an acute public health threat due to exposure to an identified hazard. RRA involves a joint assessment by the WHO country and regional offices, and headquarters. It is conducted for events with serious public health implications following pre-defined criteria within WHO. The RRA process provides a forum for the timely assessment of available data, which takes into account the contextual and hazard specific knowledge and feedback of key experts across WHO. It supports a collaborative expert prioritization of immediate actions in a time-sensitive manner. The finalized RRA report may be shared with key stakeholders that could contribute to the response. RRA reports have become well accepted documents of high practical value both within WHO and among partners.

REQUEST FOR VERIFICATION

Request for verification is defined by Article 10 of the IHR (2005) as “WHO shall request, in accordance with Article 9, verification from a State Party of reports from sources other than notifications or consultations of events which may constitute a public health emergency of international concern allegedly occurring in the State’s territory. In such cases, WHO shall inform the State Party concerned regarding the reports it is seeking to verify.”

SIGNAL

Data and/or information detected that represent a potential acute risk to human health. Signals may consist of reports of cases or deaths (individual or aggregated), potential exposure of human beings to biological, chemical or radiological and nuclear hazards, or the occurrence of natural or human-made disasters. Signals can be detected through a variety of sources (see *Detection*).

SOURCE OF INITIAL EVENT REPORTING

It is the source through which the initial information on the event was obtained. Information can be routinely obtained through NFPs or national government channels. Alternatively, WHO may receive information through detection activities (see *Detection*) as part of wider public health intelligence operations.

EXECUTIVE SUMMARY

Detecting health threats early is a crucial aspect of addressing health emergencies. Early detection plays a vital role in the comprehensive strategy of the World Health Organization (WHO) towards emergency response. Public health intelligence encompasses the identification, confirmation, analysis, evaluation, interpretation, and dissemination of information for the purpose of raising awareness, taking action and spreading knowledge.

WHO's dedicated teams at headquarters, regional, sub-regional hubs and country offices work tirelessly around the clock, every day of the year, in collaboration with national governments, and partners to carry out these public health intelligence activities.

The focus of this report is on the public health intelligence activities carried out by WHO in 2023. It emphasizes the various activities such as detection, verification, risk assessment, and dissemination of information conducted by WHO public health intelligence teams worldwide. Information on signals, verification requests, and events was gathered from different WHO databases, including, among others, the internal WHO Event Management System (EMS). Rapid Risk Assessment (RRA) reports, Event Information Site (EIS) for IHR National Focal Points, bulletins, announcements, and Disease Outbreak News (DON) reports.

Between 2004 and 2023, a total of 5910 events were reported, averaging 296 events per year. Notably, half of these events took place in both the African Region (25%) and the Region of the Americas (22%). Infectious diseases have consistently been the most common hazard type globally over the past two decades. Nevertheless, there has been a rise in the number of reported events caused by natural disasters in various WHO regions in recent years.

To improve PHI operational practices, refined signal definitions and triaging criteria were implemented in headquarters and some regions in 2023, resulting in a decrease in the number of relevant signals detected (nearly 7000 in 2022 vs. 1470 in 2023). However, an increased number of requests for verification was observed in 2023 (422), compared to 2022 (351), indicating an improved sensitivity of the detection process.

An important measure of tracking PHI efforts is timeliness, particularly in the detection, notification, verification, and information sharing. During the last five years (2019–2023), the median time to detect events was four days, the median time to notify events to WHO was two days, and the median time to respond was within a day. While timely response to event verification has been observed, improvements are needed to promptly notify events to WHO.

Overall, in 2023, 365 new events were recorded in EMS, these were either verified signals or events directly notified under the IHR (2005). Less than half of these events (43%), were initially reported by NFPs or through official national government channels, which is lower when compared to the global estimate of the previous five years.

All events recorded in EMS were risk assessed and, additionally, 52 more detailed RRA reports were produced and disseminated in 2023. Furthermore, to disseminate information to alert Member States and the wider public health community about public health events more than 125 EIS bulletins or announcements and 68 DON reports were published in 2023. In addition, WHO regions produced regular region-specific risk assessments, bulletins and situation reports.

This report highlights the importance of public health intelligence globally. It emphasizes the key role of the IHR (2005) framework for engagement by WHO with States Parties for the detection, verification, risk assessment, and information sharing regarding events that may constitute a public health emergency of international concern (PHEIC).

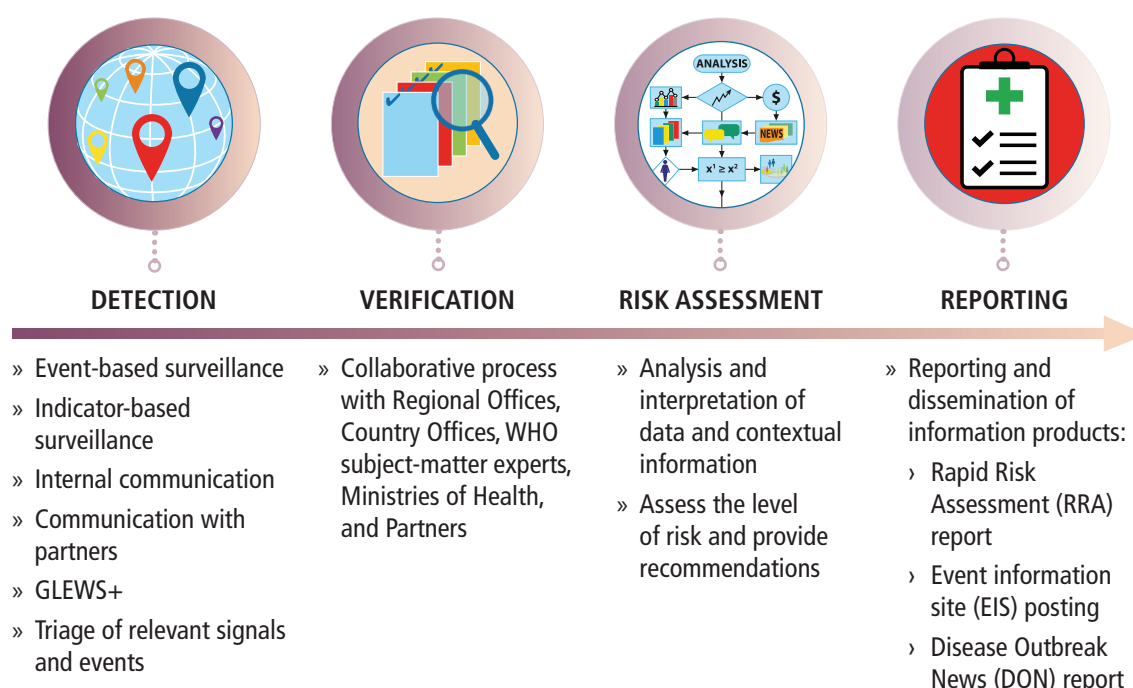
1. Background

1.1 Introduction

Early detection and rapid response are essential to tackle public health threats. The World Health Organization (WHO) has the legal mandate to safeguard health globally, and one of WHO's most important activities is the detection of and response to acute public health events.

To fulfil this mandate, WHO has implemented and continuously furthers its robust approach to public health intelligence (PHI) for the global detection, verification and risk assessment of acute public health threats. Overall, PHI is a core public health function for detecting, verifying, analysing, assessing, interpreting and generating information for action and dissemination (Fig 1).

FIG. 1 Framework of public health intelligence activities at WHO



Operationalization of PHI activities across WHO globally is underpinned by the legal mandate derived from the IHR (2005). Under the IHR (2005), which came into force in 2007, States Parties agreed to strengthen surveillance capacities, and to assess, notify and verify events that may constitute a public health emergency of international concern (PHEIC). The IHR (2005) also requires States Parties to designate a National IHR Focal Point (NFP) with whom WHO shares information through the Event Information Site for National IHR Focal Points (EIS).

Furthermore, the IHR (2005) allows WHO to disseminate information to the public through Disease Outbreak News (DON) reports.

PHI operations are conducted continuously, every day of the year, by dedicated teams at WHO headquarters and its six WHO regional offices⁴ in close collaboration with WHO country offices, national governments and partners. A full overview of States Parties — and which WHO region they are part of — is provided in Annex 1 of this report.

This report aims to provide an overview and analysis of PHI activities at WHO, with a focus on activities in 2023. The report highlights the detection, verification, risk assessment, reporting and information dissemination of activities conducted, and outputs produced by WHO PHI teams globally.

1.2 Methodology

The methodology of this report is briefly described in the following sections. The section on data sources outlines where data were obtained from and gives more details on the data sources. The section on data analysis succinctly describes how the data were analysed and which metrics and tools were utilized.

Data sources

Data on signals were obtained from different data management systems for the WHO regions and headquarters, including from the internal WHO signal management application that has been implemented since 2022. Data on the number of signals recorded between 1 January 2023 and 31 December 2023 were provided by the WHO regions, and detailed signal data detected and managed by WHO headquarters for 2023 were extracted from the signal management application.

Data on events were extracted from the WHO Event Management System (EMS). The EMS is not intended to be an exhaustive database of all acute public health events occurring worldwide. For this report, only data on events designated as substantiated acute public health events were extracted from the EMS. Events were included based on the creation date in the EMS during the past 20 years, between 1 January 2004 and 31 December 2023.

Data on requests for verification for select signals were obtained through the WHO regional offices. This was tracked by the monitoring of requests for verification sent by the WHO regional offices to NFPs and WHO country offices and their respective responses between

4. There is a World Health Organization (WHO) regional office in each of the six WHO regions: the WHO African Region (AFR), the WHO Region of the Americas (AMR), the WHO Eastern Mediterranean Region (EMR), the WHO European Region (EUR), the WHO South-East Asia Region (SEAR), and the WHO Western Pacific Region (WPR).

1 January 2020 and 31 December 2023. The timeliness of responses was reviewed by using a benchmark of 48 hours after the request was sent.

Data on EIS bulletins and announcements were extracted from the EIS platform. Information on EIS bulletins and announcements that were posted between 1 January 2019 and 31 December 2023 were extracted. Similarly, data on Rapid Risk Assessment (RRA) reports were extracted from the EMS into a bespoke database for the report, with information extracted for RRA reports published between 1 January 2019 and 31 December 2023. Data on DON reports were extracted from an internal WHO database, with information extracted for DON reports published between 1 January 2019 and 31 December 2023.

Data analysis

The main findings were assessed for signals, request for verification, events, RRA reports, EIS bulletins and announcements, and DON reports. Findings overall were stratified by year over multi-year periods, with a particular focus on 2023, and by WHO region. In addition, for events, results were assessed by the channel through which the event was initially reported (either the NFP or national government channels versus WHO or other sources, simply referred to as WHO), type of disease, condition or hazard⁵. Similarly, findings for RRA reports, EIS bulletins and DON reports were analysed by disease, condition or hazard.

Measures of timeliness for events that were published in the EIS were assessed using the three components of the detect and respond indicator⁶: *time to detect*, *time to notify*, and *time to respond*. These components use approximate event milestones (date event started, date event detected, date event notified to WHO, and date response measures were implemented) to estimate the median and interquartile (IQR) ranges, in days. Negative values (such as if the response started prior to notification to WHO) were set to zero. The *time to detect* is calculated as the time between the date event started and date event detected; the *time to notify* is calculated as the time between the date event detected and the date event notified to WHO; and the *time to respond* is calculated as the time between the date event detected and the date response measures were implemented.

Data were presented as averages, medians or aggregated. Tables and figures were created using R (version 4.0.3, R Foundation for Statistical Computing, Vienna, Austria) and RStudio (version 1.3.1093, Boston, MA, USA). Previous and future reports may show minor differences due to routine updates and cleaning of signal, request for verification, EMS, RRA, EIS and DON data.

5. During 2022, select hazard types were reclassified in the EMS. The following hazard classifications that were used in previous reports (animal, zoonosis, and food safety) were reclassified to other respective hazards, and therefore, are no longer shown as separate classifications.

6. Additional information regarding the calculation methods and limitations for this indicator are available from the Thirteenth General Programme of Work (GPW13): methods for impact measurement. Geneva: World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO

1.3 Surveillance for public health intelligence

Public health emergencies are occurring with increasing frequency and complexity, exposing weaknesses in current surveillance efforts for informing decision-making at all levels. Effective national surveillance is the foundation of global public health security, as all activities to prevent and mitigate the impact of epidemics, pandemics and other emergencies depend on our collective surveillance capabilities.

At the Seventy-fifth World Health Assembly in May 2022, the WHO Director-General presented a harmonizing framework to strengthen the global architecture for health emergency preparedness, response, and resilience (HEPR). This framework builds upon more than 300 recommendations from various independent reviews of the global response to COVID-19 and reports on previous outbreaks as well as the views of Member States and ongoing multilateral consultation processes. Under the proposed global architecture, the ability to effectively prevent, prepare for, detect, respond to and recover from health emergencies at subnational, national, regional and global levels depends on operational readiness and capacities in five interconnected components:

1. Collaborative surveillance
2. Community protection
3. Safe and scalable clinical care
4. Access to countermeasures
5. Emergency coordination

Collaborative surveillance is defined as the systematic strengthening of capacity and collaboration among diverse stakeholders, both within and beyond the health sector, with the ultimate goal of enhancing PHI and improving evidence for decision-making.

The collaborative surveillance concept builds upon robust routine public health surveillance, health systems monitoring, and laboratory surveillance, while drawing insights from other data sources and applying advanced data and analytical approaches to enable the generation of contextualized insights on hazards, threats and risks, populations affected, and their contexts.

Collaborative surveillance reinforces the need for the triangulation of multisource information for risk assessment and to inform decision making.

In the mosaic of surveillance systems that builds collaborative surveillance, PHI activities play a strategic role in identifying, collecting, connecting, synthesizing, analysing, assessing, interpreting, and generating a wide range of information for actionable insights and disseminating these for informed and effective decision-making to protect and improve the health of populations.

Operationalization of PHI activities across WHO globally is underpinned by the legal mandate derived from the IHR (2005). PHI activities at WHO are conducted continuously throughout every day of the year to systematically detect, verify and respond to acute public health threats around the world. Specifically, it involves four interlinked steps to guide public health action and decision-making: detection, verification, risk assessment, and reporting and dissemination.

Traditional surveillance approaches, such as monitoring the number of cases and deaths, are insufficient for the management of complex public health threats. As such, WHO has developed a new model for the surveillance of emerging threats, called *pandemic and epidemic intelligence*, that builds upon a range of traditional approaches⁷ and non-traditional approaches⁸.

To capture signals of potential public health events, WHO uses a variety of information channels. One of the main channels currently used is the Epidemic Intelligence from Open Sources (EIOS) system, which processes information from online news and social media, government and official websites, and blogs and expert groups, and applies natural language processing tools that facilitate screening of information for signals of interest. A second information channel is facilitated by dedicated WHO email accounts, for direct communications, including information alerts and information sharing from partners and WHO colleagues, such as field investigation reports or data, including from indicator-based surveillance. A third information channel is GLEWS+, a global early warning system for health threats and emerging risks at the human-animal interface, maintained jointly by the Food and Agriculture Organization (FAO), WHO, and the World Organisation for Animal Health (WOAH).

Key regional surveillance sources

In addition to the surveillance sources described above, regional specific surveillance systems and mechanisms exist which feed into the public health intelligence processes. These are described below.

In the WHO African Region, Integrated Disease Surveillance and Response (IDSR) has been the primary source for surveillance. The IDSR ensures the availability of guidelines for integrated disease surveillance and response. It provides the necessary technical support for strengthening national communicable disease surveillance systems through the implementation of the regional strategy for integrated disease surveillance and the IHR 2005. Additionally, the Region is utilizing the EIOS to detect signals of public threats published online. In 2023, the EIOS initiative was expanded to 16 additional Member States of the Region, bringing the total to 33 Member States in the Region.

In the WHO South-East Asia Region, EIOS is used as the primary source for the regional event-based surveillance to screen signals, while it is complemented by additional sources such as HealthMap, ProMed, GPHIN and GDACS. Information is also gathered from official websites and social media of the Ministries of Health and relevant institutions.

7. Example of traditional approaches: human patients who present at healthcare services; hospitalized patients; laboratory diagnoses; pathogen genome sequencing; mortality estimation; on-the-ground investigations, including active case finding and contact tracing.

8. Example of non-traditional approaches: detection of signals of public health events from unstructured textual online; information, including social media channels; data from sectors not routinely included in public health surveillance, such as animal health, occupational health and police reports; community reporting of cases, deaths or events among humans or animals; geospatial, remote-sensing and mobile-phone mobility data; crowdsourcing open-source case-based data.

In the WHO Western Pacific Region, Pacific Island countries and areas (PICs) contribute weekly to the Pacific Syndromic Surveillance report through the Early Warning Alert and Response System. Influenza-like illness and dengue-like illness data from PICs are used in the regional bi-weekly seasonal influenza and dengue reports.

In the WHO Eastern Mediterranean Region besides the EIOS, innovative tools are used for social media screening, EWARS is already implemented in some countries of the Region, in addition to Integrated Disease Surveillance (IDS), which was endorsed by the Regional Committee in 2021.

In the Americas Region, public health intelligence is conducted with a One Health approach, fostering collaborative surveillance at different levels. The signals captured by the EIOS system as well as from other sources are structurally and periodically monitored. Close collaboration with national authorities engaged in epidemic intelligence activities ensures direct notifications regarding event-based and indicator-based surveillance. Coordination with internal alert and rapid response networks operated by WHO/PAHO, such as the International Food Safety Authorities Network (INFOSAN) and the WHO Global Surveillance and Monitoring System for Medical Products, is maintained, along with external partners and their surveillance systems, such as the World Animal Health Information System (WAHIS) and with the International Atomic Energy Agency (IAEA) to ensure a One Health approach.

WHO's Early Warning, Alert and Response System (EWARS)

To facilitate early detection of disease outbreaks in fragile, conflict-affected and vulnerable contexts and countries experiencing health emergencies, WHO advocates for the implementation of an Early Warning, Alert and Response System. Rapid identification of disease outbreaks is crucial for establishing an efficient public health response to mitigate severe consequences and further spread of diseases. The Early warning alert and response (EWAR) in emergencies: an operational guidance was published in January 2023. Through its application, the operational guidance aims to contribute to:

- » earlier detection of acute public health events
- » earlier and more effective response
- » reduced impact of emergencies on health
- » increased trust of the population in the (public) health system
- » fulfilling our collective commitments to the International Health Regulations (IHR, 2005).

In 2015 WHO developed a tool to assist with implementing an Early Warning, Alert and Response System (EWARS). EWARS Mobile, formerly called EWARS in a box is a simple and cost-effective tool which enables rapid electronic reporting of epidemic-prone diseases, and integrates easy and automated visualizations to support decision making in disease surveillance operations by combining many surveillance mechanisms into a defined set of indicators.

EWARS Mobile provides a comprehensive package of services enabling real-time mobile reporting on epidemic-prone diseases from the most difficult and remote field setting, even in the absence of reliable internet connection or electricity supply. It supports countries to implement surveillance activities (indicator-based surveillance, event-based surveillance, community-based surveillance, case-based information and line lists, laboratory surveillance, etc.) as part of preparation for and response to emergencies (disease outbreak, natural disaster, conflicts, etc.) and where needed with interactive alert management.

EWARS Mobile services are rapidly deployable (within 48 hours) and scalable to support emergency response and fragile states and can also be integrated or interoperated with routine surveillance tools to complement it. EWARS reporting sites can range from primary health care facilities in emergency-affected or besieged areas, camp clinics, mobile clinics, field laboratories to the community. Reporting can be done by anyone, including medical doctors, nurses, midwives working in a camp for internally displaced persons or in local primary health care centre, community health workers, community volunteers or community members themselves.

In 2023, over 100 million people across 36 countries were covered with surveillance activities implemented through EWARS Mobile, an essential tool in the public health intelligence sphere.

2. Main findings

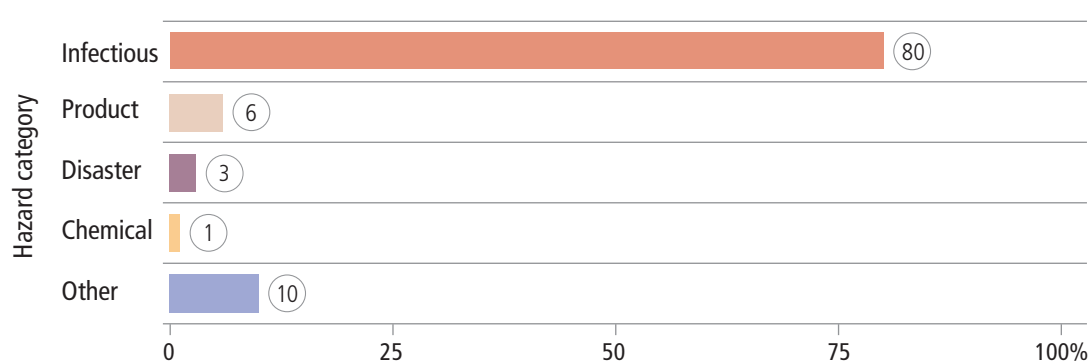
2.1 Detection, verification and bilateral communication for acute public health events

Detection

In 2023, 1470 signals of potential public health threats were detected by dedicated teams at WHO headquarters and all regional offices except for the Western Pacific Regional Office. This is a marked decrease compared to the signals reported in 2022 ($n = 6855$), and is due to the refinement of signal definitions and triaging criteria used in 2023 to better align with operational practices.

Among the 533 signals detected by WHO headquarters in 2023, the majority were due to infectious hazards (80%, 425/533), followed by product-related hazards (6%, 32/533), disasters (3% 17/533), or chemical hazards (1%, 4/533). The remaining signals were the result of other hazard categories (i.e. “undetermined” and “societal”) (10%, 55/533) (Fig. 2).

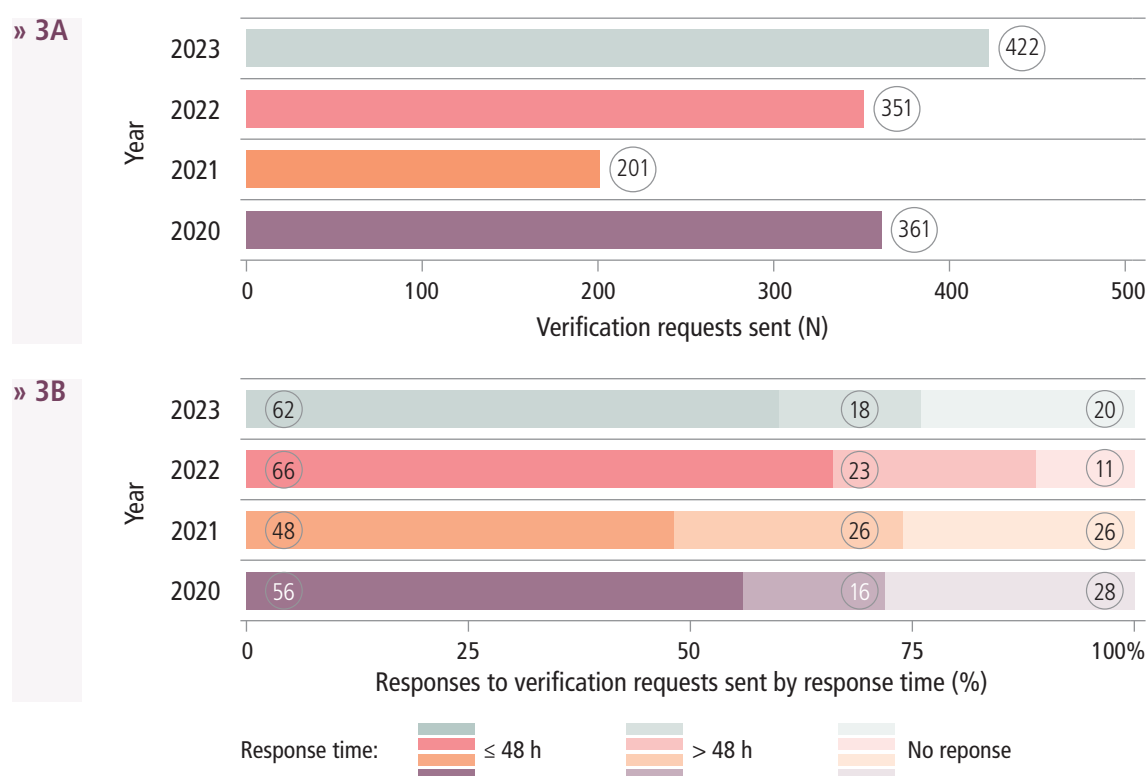
FIG. 2 Percentage of signals detected at WHO headquarters, by hazard category, 2023



Verification of signals

Detected signals are triangulated and triaged by PHI analysts, a group of epidemiologists with multidisciplinary backgrounds. After assessment, verification requests are sent for those signals for which verification by NFPs or others was required. In 2023, 422 requests for verification were sent by WHO regional and country offices. This is an increase of 20% compared to 2022 and larger than the previous highest number of requests sent, in 2020 (Fig. 3).

FIG. 3 (A) Number of verification requests and (B) percentage of responses to verification requests sent by response time, 2020–2023



In one WHO region, timeliness of the response to the verification requests sent in 2023 was only available for a subset of the verification requests. Therefore, to assess the response timeliness, only 359 requests for verification were included. For this subset, a response was received for the majority (76%, 273/359) in 2023. Among those for which a response was received, 225 (62%) were received within 48 hours of the request being sent. Compared to 2022, the overall response rate decreased by 9% and the response rate for those received within 48 hours decreased by 6%. In addition, the percentage of requests for verification for which no response was received has doubled compared to 2022, but is comparable to the percentage during the period of 2020–2021 (Fig. 3B).

The number of requests for verification sent differed between WHO regions. This is in part due to regional differences in the underlying health threats that occur and partly due to procedural and information management practices. In 2023, there was an improved overall response rate, within 48 hours, for all requests for verification sent, compared to the previous year, for half of the WHO regions (Fig. A1).

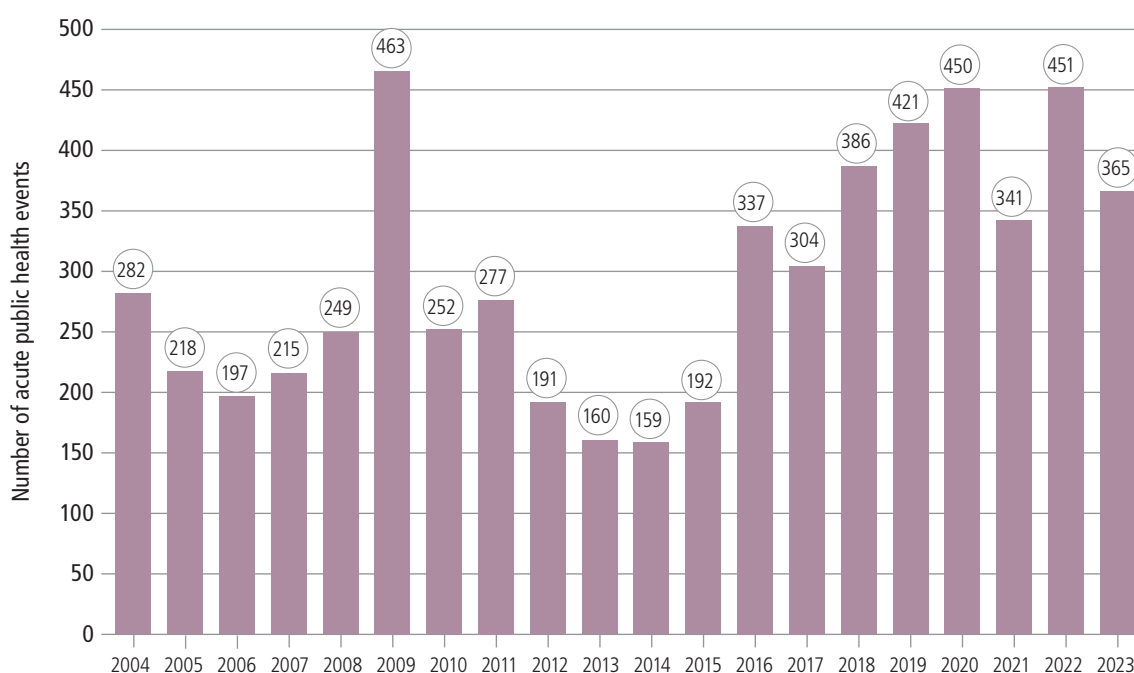
2.2 Acute public health events

In this section, all the findings refer to events that have been verified and designated as substantiated, which are events where the presence of a hazard is confirmed, or the number of human cases exceeds established thresholds. To facilitate readability, they are referred to as “acute public health events” or simply “events”, rather than “verified and substantiated acute public health events”.

Acute public health events by year and WHO region

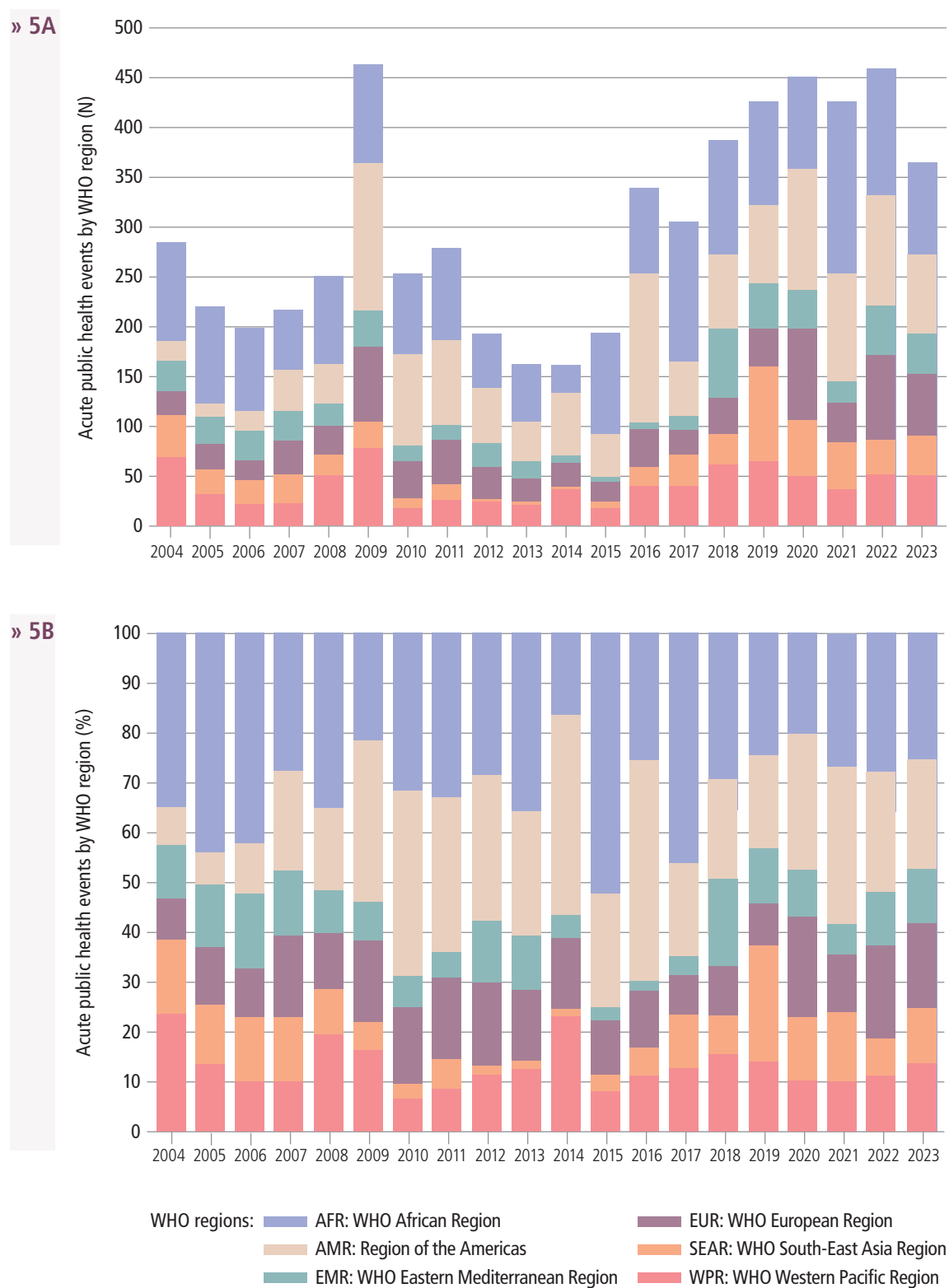
In 2023, there were 365 acute public health events reported globally. This is a decrease compared to the previous year, but in line with the number of events reported in recent years (Fig. 4). In the past 20 years (2004–2023), 5910 events overall were reported, and the number of events per year ranged from 159 to 463, with a median of 280 events.

FIG. 4 Acute public health events, in absolute numbers, 2004–2023 (N = 5910)



The number of reported acute public health events varied by WHO region. In 2023, the highest number of events were reported in the African Region (25%, 93) and the Region of the Americas (22%, 80). The European Region reported 17% of events (62) followed by the Western Pacific Region (14%, 50), and the Eastern Mediterranean and South-East Asia Regions, which each reported 40 (11%) events (Fig. 5). In the past 20 years, the largest number of events reported have consistently been from the African Region or the Region of the Americas. In recent years, events reported from these two regions individually have accounted for approximately 20–30% of all events, while the events in each of the other WHO regions generally have ranged from 5% to 20% (Fig. 5).

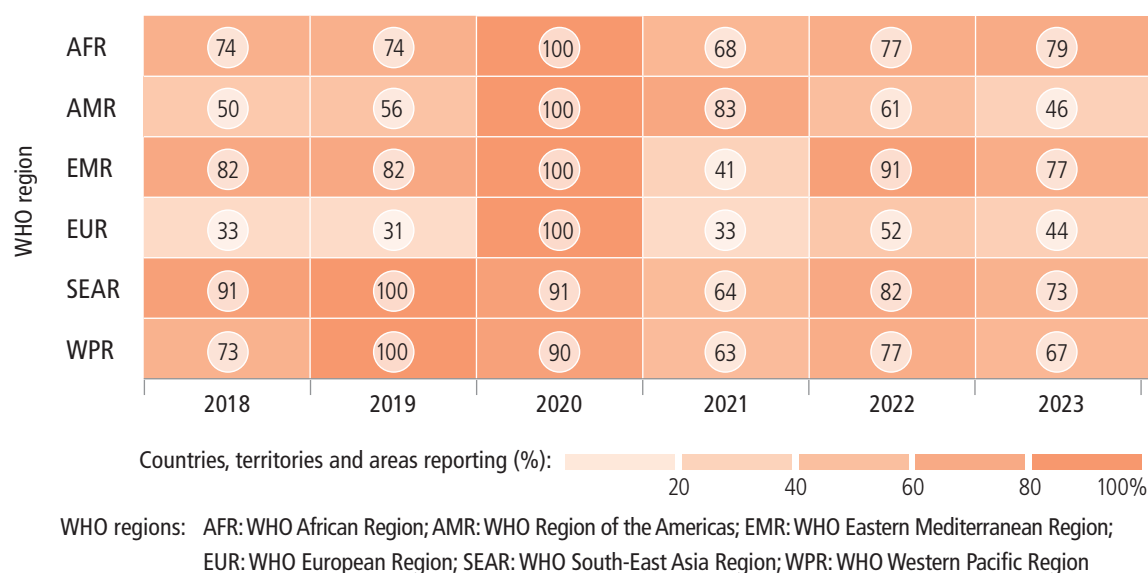
FIG. 5 Acute public health events in (A) absolute number and (B) percentage, by WHO region, 2004–2023 (N = 5910)



Various factors influence the differences in events reported by WHO region, including the number of countries, territories and areas per WHO region. When assessing the proportion of countries, territories and areas reporting acute public health events by year, in the last six years, most of the WHO regions had at least two-thirds of countries, territories and areas reporting at least one acute public health event (Fig. 6).

In 2023, four WHO regions had at least two-thirds of countries, territories and areas reporting an event, with no distinctive pattern related to the size of the region or number of events. Importantly, this only reflects newly reported acute public health events and does not provide any information on ongoing public health events that continue to be monitored.

FIG. 6 Percentage of countries, territories and areas reporting at least one acute public health event, by WHO region, 2018–2023



Initial source of event reporting

In 2023, less than half of all events (43%, 157/365) were initially reported by NFPs or through official national government channels, which is lower when compared to the global estimate of the previous five years (46%, 931/2028).

Between 2019 and 2023, 58–73% of acute public health events were initially reported by NFPs or through national governments channels in the Region of the Americas, the European Region and the Western Pacific Region. On the other hand, for the other three WHO regions, the majority of events were initially identified by WHO. Between 9–35% of events in the African, Eastern Mediterranean and South-East Asia Regions were initially reported by NFPs or through national governments channels (Fig. 7).

In 2023, the number of events reported by the NFP or national government agency decreased or remained the same compared to the previous year for all WHO regions (Fig. A2). However, looking at longer-term trends and comparing five-year periods within each WHO region

during the past 20 years, there was an increasing trend in the NFP and national government agency as the initial source of events in the Region of the Americas, Eastern Mediterranean Region, European Region and Western Pacific Region (Fig. 8).

FIG. 7 Initial source of information for events, by WHO region, 2019–2023

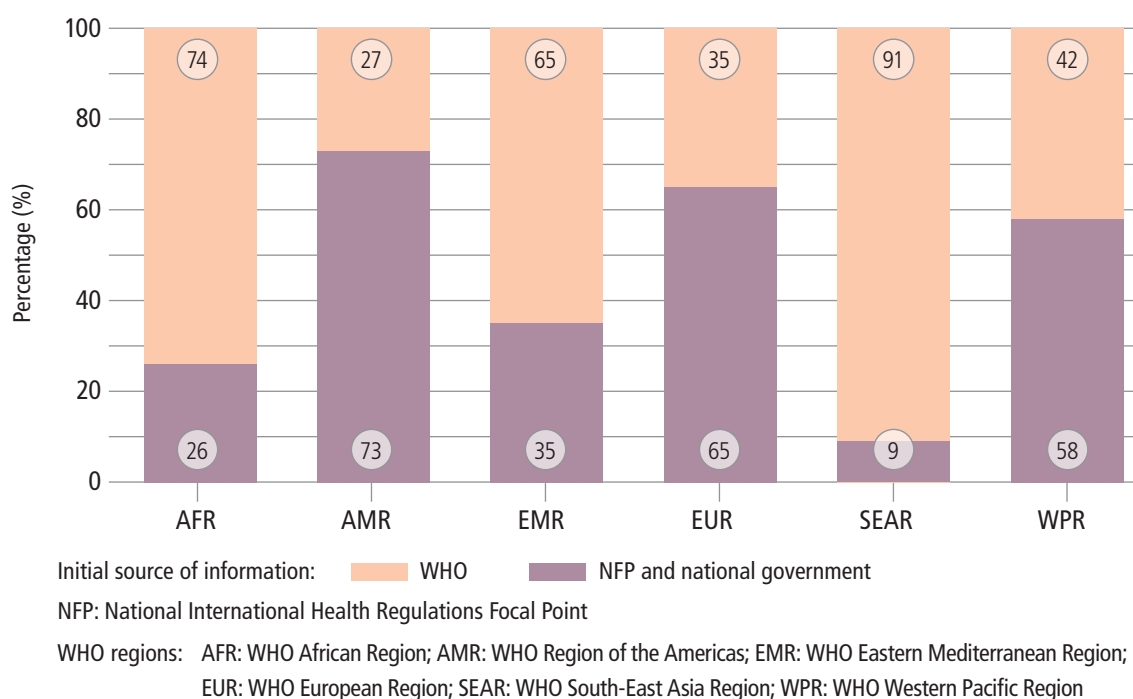
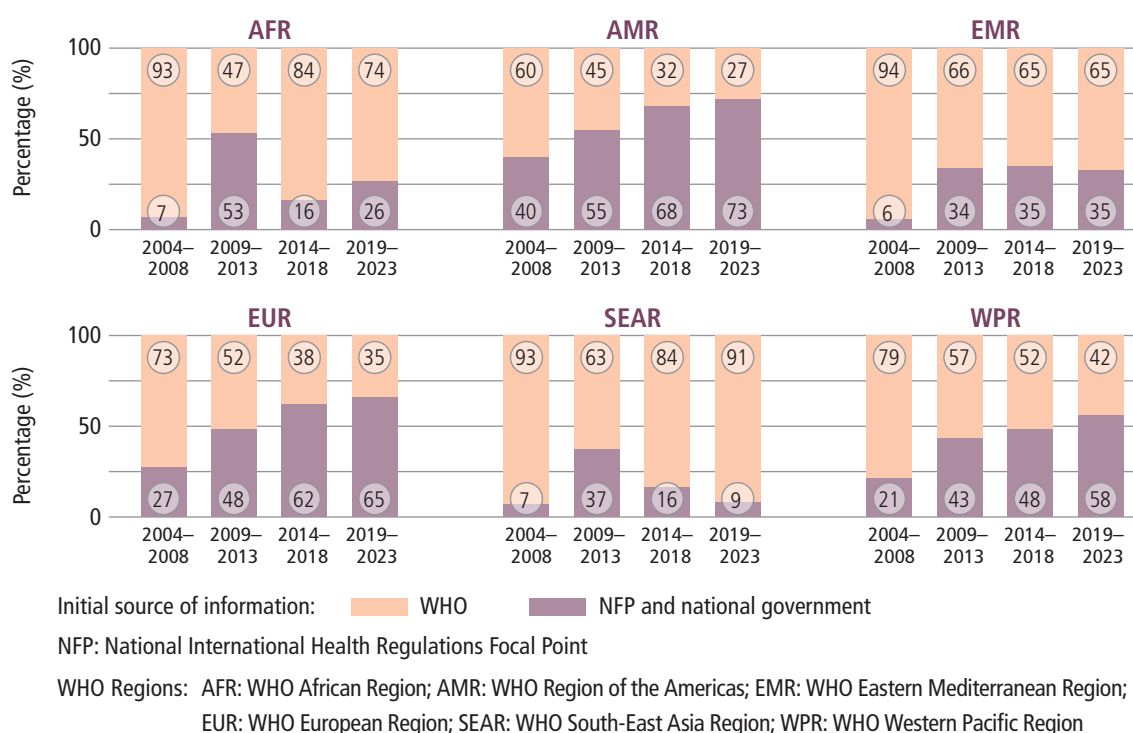


FIG. 8 Initial source of information for events, by WHO region, for each five-year period, 2004–2023



Hazard type of acute public health events

During the past 20 years (2004–2023), infectious diseases have remained the predominant hazard type for events globally (Fig. 9). In 2023, infectious diseases accounted for 79% (289) of all events globally, while the second most common cause were disasters (10%, 36). The remaining events were caused by chemical, product or other hazards, ranging from 2% to 10% per hazard. Of note, beginning in 2022, food safety and animal/zoonosis hazards were reclassified to other respective hazards, and therefore, explain the absence of these hazard types in 2023.

The main hazard type of acute public health events differed by WHO region (Table A1, Fig. A3, A4). Most notably, events caused by disasters have been more frequently recorded, particularly in the previous five years, in the African, Eastern Mediterranean, South-East Asia and Western Pacific Regions, whereas in the Region of the Americas and the European Region, events due to product-related hazards have been more frequently recorded in recent years. Additionally, while only accounting for a small proportion, events due to chemical-related hazards increased in 2023 compared to previous years.

To examine the most common diseases or conditions of an infectious nature, the ten most frequently occurring diseases or conditions of an infectious nature were identified, for each year of the past five years and across the five-year period 2019–2023 (Table 1).

Over the five-year period there were 25 different diseases or conditions that were among the ten most common causes for acute public health events of an infectious nature in a given year. Five diseases (cholera, dengue fever, influenza, measles and poliomyelitis) were among the ten most frequent every year (Table 1). Aggregated across the five-year period, infectious disease events were dominated by coronavirus disease (COVID-19), followed by dengue fever, cholera, measles and mpox.

In 2023, most acute public health events of an infectious nature were largely due to dengue fever, followed by measles, cholera, influenza (due to identified avian or animal influenza virus) and poliomyelitis. Additional frequently reported infectious diseases, when examining events by WHO region for 2023, include yellow fever, malaria, legionellosis, anthrax, meningococcal disease, salmonella infections, diphtheria and Zika virus disease (Table A3).

FIG. 9 Acute public health events by hazard type: (A) in absolute number and (B) in percentage, 2004–2023, (N = 5910)



TABLE 1 Acute public health events for the most frequently reported diseases, conditions or hazards of an infectious nature, for each year of the past five years and across the five-year period, 2019–2023

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	—	—	—	11	—	—
2	Anthrax	6	—	—	—	9	—
3	Chikungunya	9	—	—	—	—	—
4	Cholera	23	6	20	43	19	111
5	COVID-19	—	226	44	—	—	278
6	Crimean–Congo haemorrhagic fever	9	—	—	12	8	36
7	Dengue fever	42	16	15	23	33	129
8	Diphtheria	—	6	—	11	12	35
9	Gastroenteritis	6	—	—	—	—	—
10	Influenza due to identified avian or animal influenza virus	—	6	17	17	18	62
11	Influenza due to identified human influenza virus	12	—	—	—	—	—
12	Lassa fever	7	—	5	—	—	—
13	Malaria	—	8	5	—	—	—
14	Measles	45	8	11	23	21	108
15	Meningitis	—	—	—	13	—	—
16	Mpox	—	—	—	90	10	100
17	Multisystem inflammatory syndrome in children (MIS–C)	—	18	8	—	—	—
18	Poliomyelitis	21	8	19	13	16	77
19	Yellow fever	—	14	10	—	8	46

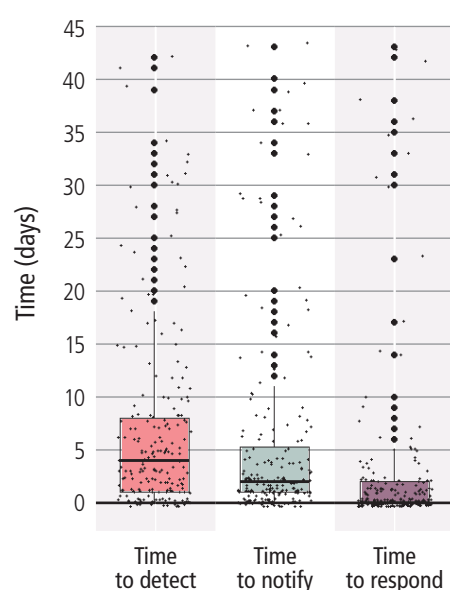
Note: This table indicates the number of events for the ten most frequently occurring diseases or conditions of an infectious nature, per year or across the five-year period. Therefore, each column (one for each year or one indicating the five-year period) will only contain values for ten diseases, conditions or hazards.

Timeliness of events published in the Event Information Site (EIS): detection, notification, and response

An important measure of tracking PHI efforts is timeliness, particularly in the detection, notification and response to acute public health events. During the past five years (2019–2023), there were 300 events reported by all WHO regions that had an Event Information Site (EIS) bulletin published and had sufficient information available to approximate the detect and respond indicator's three components: *time to detect*, *time to notify* and *time to respond*⁹.

During this five-year period, the median “time to detect” events was 4 days (IQR range 1–8), the median “time to notify” events to WHO was 2 days (IQR 1–5.25), and the median “time to respond” (from the date the event was detected) was 0 days (IQR 0–2 days) (Fig. 10).

FIG. 10 Timeliness of events published in the Event Information Site, in days, for detection, notification and response, 2019–2023

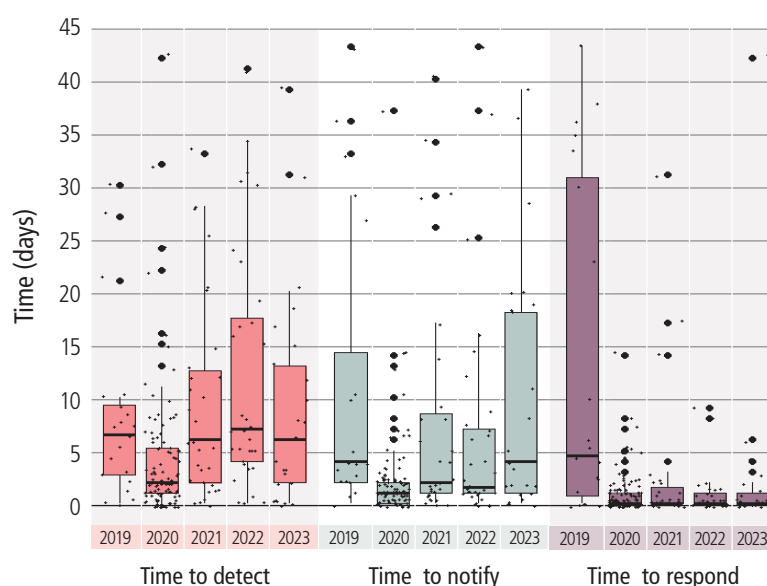


Comparing each year during the five-year period (2019–2023), these three measures were the timeliest (lowest) during 2020, likely related to the COVID-19 pandemic (Fig. 11). The median time to detect was 6–7 days during the remaining years. The median and upper quartiles for time to notify remained lower during 2021–2022, and were higher in 2019 and 2023. The time to respond was highest in 2019 compared to all remaining years, after which they were comparable each year, suggesting a substantial improvement in this measure.

During the combined five-year period, all WHO regions generally had a median time to detect of 10 days or less, with the median time to notify and time to respond was lower (less than 5 days); however, greater variation was observed across the WHO regions when comparing these measures annually (Fig. A5, A6).

9. Additional information regarding the calculation methods and limitations for this indicator are available from the Thirteenth General Programme of Work (GPW13): methods for impact measurement. Geneva: World Health Organization; 2020. Licence: CC BY-NC-SA 3.0 IGO.

FIG. 11 Timeliness of events published in the Event Information Site, in days, for detection, notification, and response for each year of the past five years, 2019–2023



2.3 Risk assessment of acute public health events

Following early detection of an event a rapid assessment is critical to prevent public health events from becoming emergencies and to inform decision-making for an effective response.

Within WHO the most important assessments for guiding initial emergency response are the RRAs conducted for acute public health events, and the Public Health Situation Analysis (PHSA) conducted to determine the immediate needs of a population following sudden onset emergencies or deterioration of humanitarian crises.

RRAs are conducted for large scale acute events potentially exceeding the capacity of the affected country to respond, for acute events with international implications and for those requiring response and coordination support by WHO. The RRA provides information to understand and communicate the public health risk(s) and the potential requirements for a response by WHO and partners. The main objectives of the WHO RRA are to characterize the risk to public health and to recommend the most effective public health actions that will prevent any further amplification of the event assessed.

The outputs of the RRA formalize the WHO's assessment of risks and reflect WHO's independent assessment including country's capacities and vulnerabilities to control an outbreak and highlight urgent actions required such as refer an event for review of IHR Emergency Committee for consideration as a PHEIC, activation of Emergency Response Framework mechanisms with

recommendation of setting up a grading discussion (an internal WHO activation procedure where the grade indicates the level of operational response required by WHO).

In addition to the standard RRA reports performed for an event occurring in one single country, there are regional and global RRA reports for a single acute public health event involving several countries in one WHO region or across the six regions.

For humanitarian emergencies such as natural disasters and conflicts, in which multiple concurrent health threats are present (e.g., trauma, infectious diseases, exposure to chemicals and deterioration in access to health services), WHO's Health Emergencies Programme regularly undertakes a similar but distinct process, culminating in a PHSA¹⁰.

A PHSA is conducted in response to a sudden-onset emergency or deterioration in a protracted crisis. It is one of the key outputs of WHO and/or the Health Cluster as stipulated in the Public Health Information Services Standards. The main objectives of the PHSA are to characterize the major needs and risks faced by a population, including elements of humanitarian response capacity. It is a holistic assessment of all the public health issues and response landscape in a given context, and is most applicable in settings with activated health coordination mechanisms (e.g. Health Clusters). The PHSA does not include recommendations about priority interventions, but rather functions as a platform for joint planning with partners. It provides a comprehensive analysis of the existing health status and potential health threats that the population may face over time, the functioning of the health system and humanitarian health system performance. It aims to provide all health sector partners, including local and national authorities, NGOs, donor agencies and United Nations agencies with a common understanding of the public health situation in an emergency in order to inform evidence-based collective humanitarian health response planning.

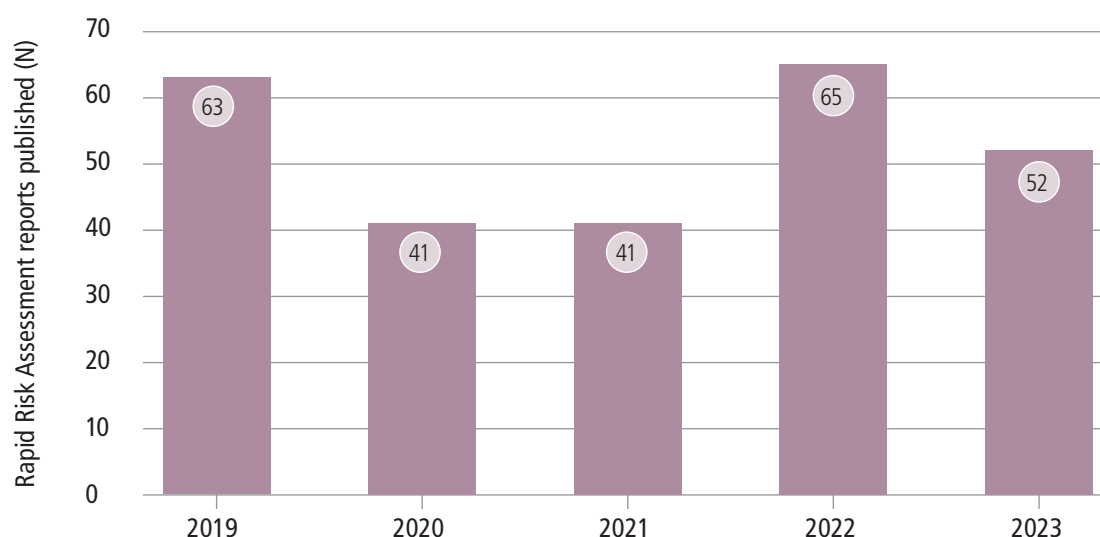
In 2023, PHSAs were conducted and/or updated for major emergencies, including the war in Ukraine; the refugee crisis resulting from the Ukraine war; the complex emergency in Syria; the drought and food insecurity crisis in the Horn of Africa; the complex emergency in the Sahel region; floods in Pakistan; intercommunal violence in Sudan; a tropical storm in Malawi; and floods in Madagascar.

In subsequent sections, only information on RRAs will be presented and it is important to note that these risk assessments are only a very selected subset of those that WHO conducted.

Rapid Risk Assessment reports, by year and WHO region

Between 2019 and 2023, 262 RRA reports were published, ranging from 41 to 65 per year. The average number of RRA reports published in the past five years was 52 per year; during the first two years of the COVID-19 pandemic, 2020 and 2021, there was a decrease in the overall number of RRA reports published (Fig. 12).

10. More information about the Public Health Situation Analysis can be found at: <https://healthcluster.who.int/our-work/task-teams/information-management-task-team/public-health-information-services-toolkit>. Examples can be found at: <https://response.reliefweb.int/ukraine>.

FIG. 12 Number of Rapid Risk Assessment reports published by year, 2019–2023 in absolute number

The majority of RRA reports published during the past five years were related to acute public health events in the African Region (49%, 128/262), followed by the Eastern Mediterranean Region (15%, 38/262) and by the South-East Asia Region (10%, 27/262). In addition, a marked increase in multi-country RRAs was seen with 10% (26/262) of RRA reports addressing global acute public health events, while 1% (3/262) were related to multi-regional events, simultaneously affecting two WHO regions (Table 2).

Though the absolute number of RRAs decreased during 2020–2021 there was an observed shift towards the type of RRAs performed, as the format of the RRA report was adapted to assess a regional event (one single region with multiple countries assessed) or global risk of an event. A total of 26 global RRA reports were published in the period 2020–2023, likely reflecting the trend of geographical expansion of certain infectious hazards (vector borne and vaccine preventable disease) and the need of an assessment of the risk at global level (Table 3).

Rapid Risk Assessment reports by hazard, disease or condition

During the 2019–2023 period, 99% (259/262) of the RRAs performed were related to an infectious hazard. The remaining 1% (3/262) involved acute public health events for environmental contamination ($n = 1$) and the risk of a biological hazard from a seized laboratory ($n = 2$).

TABLE 2 Rapid Risk Assessment reports published by WHO region, 2019–2023, in absolute number and percentage

WHO Region	2019 N = 63	2020 N = 41	2021 N = 41	2022 N = 65	2023 N = 52	2019–2023 N = 262
AFR	29 (46%)	26 (63%)	22 (54%)	30 (46%)	21 (40%)	128 (49%)
AMR**	7 (11%)	0 (0%)	1 (2%)	3 (5%)	4 (8%)	15 (6%)
EMR	12 (19%)	0 (0%)	6 (15%)	14 (22%)	6 (12%)	38 (15%)
EUR	4 (6%)	3 (7%)	2 (5%)	1 (2%)	2 (4%)	12 (5%)
SEAR	7 (11%)	2 (5%)	4 (10%)	5 (8%)	9 (17%)	27 (10%)
WPR	4 (6%)	7 (17%)	0 (0%)	1 (2%)	1 (2%)	13 (5%)
AFR & EMR	0 (0%)	0 (0%)	1 (2%)	0 (0%)	0 (0%)	1 (0%)
AMR & EUR	0 (0%)	0 (0%)	0 (0%)	2 (3%)	0 (0%)	2 (1%)
Global	0 (0%)	3 (7%)	5 (12%)	9 (14%)	9 (17%)	26 (10%)

** Rapid Risk Assessment reports from the WHO Regional Office for the Americas

WHO Regions: AFR: WHO African Region; AMR: WHO Region of the Americas; EMR: WHO Eastern Mediterranean Region;
EUR: WHO European Region; SEAR: WHO South-East Asia Region; WPR: WHO Western Pacific Region

The five most common diseases or conditions for which RRA reports were published between 2019 and 2023 were cholera (16%, 42/262), measles (10%, 25/262), COVID-19 (9%, 24/262), Ebola virus disease (9%, 23/262) and dengue fever (7%, 19/262). These five diseases resulted in 51% (133/262) of all RRA reports produced in this time period. Cholera was the only disease for which at least one RRA report was produced each year between 2019 and 2023, and there were six diseases (measles, COVID-19, Ebola virus disease, dengue fever, poliomyelitis and Lassa fever, with at least one RRA report for four of the five years).

By WHO region the disease event with the most published RRA reports was cholera in the African and Eastern Mediterranean Regions, vector-borne diseases, such as yellow fever in the Region of the Americas, Nipah virus infection and dengue fever in the South-East Asia Region and COVID-19 in the Western Pacific Region. In the European Region, most RRA reports were published for COVID-19, antibiotic resistant bacteria and poliomyelitis. (Tables A2, A5–A10). Global RRA reports were first published in 2020 for COVID-19 and in 2022 for cholera and mpox (Table A4).

TABLE 3 Rapid Risk Assessment reports published by disease, condition or hazard, 2019–2023

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	–	–	–	1	–	1
2	Acute kidney injury	–	–	–	1	–	1
3	Acute watery diarrhoea	1	–	–	–	–	1
4	Anthrax	–	–	–	–	2	2
5	Antibiotic-resistant bacterial infection	2	–	–	1	–	3
6	Argentine haemorrhagic fever	–	1	–	–	–	1
7	Chikungunya	2	1	–	–	1	4
8	Cholera	6	3	8	19	6	42
9	COVID-19	–	11	5	5	3	24
10	Crimean–Congo haemorrhagic fever	–	1	–	1	1	3
11	Dengue fever	6	–	2	4	7	19
12	Diphtheria	–	1	–	–	4	5
13	Ebola virus disease	11	6	4	2	–	23
14	Environmental contamination	–	1	–	–	–	1
15	Hantavirus pulmonary syndrome	1	–	–	–	–	1
16	Hepatitis E	–	2	2	–	–	4
17	HIV infection	1	–	1	–	–	2
18	Infectious acute-febrile syndrome	–	–	1	–	–	1
19	Influenza, avian/animal virus	–	2	1	–	3	6
20	Intestinal infectious diseases	–	–	–	–	1	1
21	Japanese encephalitis	–	–	–	1	–	1
22	Lassa fever	3	1	–	3	1	8
23	Leptospirosis	–	–	–	1	–	1
24	Malaria	3	–	2	–	1	6
25	Marburg virus disease	–	–	1	1	3	5
26	Measles	9	1	–	9	6	25
27	Meningococcal disease/Meningitis	1	–	1	4	3	9
28	MERS-CoV	–	–	–	–	1	1
29	Mpox	–	1	–	4	2	7
30	Nipah virus infection	2	–	1	–	2	5
31	Plague	–	1	2	–	–	3
32	Poliomyelitis	6	–	2	2	2	12
33	Rabies	–	–	–	–	1	1
34	Rift Valley Fever	1	1	–	1	–	3

TABLE 3 Rapid Risk Assessment reports published by disease, condition or hazard, 2019–2023, (continued)

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
35	Risk of biological hazard	–	–	–	–	2	2
36	<i>Salmonella typhimurium</i> infection	–	–	–	1	–	1
37	Smallpox vaccine	1	–	–	–	–	1
38	Sudan virus disease	–	–	–	2	–	2
39	Viral haemorrhagic fever	–	1	–	–	–	1
40	Whooping cough (pertussis)	1	–	–	–	–	1
41	Yellow fever	5	6	5	2	–	18
42	Zika virus disease	1	–	2	–	–	3

Rapid Risk Assessment reports by type and risk level

The majority (83%, 218/262) of RRA reports published within the reporting period concerned acute public health events in a single country. The remaining RRA reports were for global (10%, 26/262), multi-regional (1%, 3/262) or regional (6%, 15/262) acute public health events. RRA reports assess the risk at various geographical scales. For example, a single country RRA reports will assess the risk at the country, regional and global levels, while regional RRA reports will assess the risk at regional and global levels, and global RRA reports only assess the risk at a global scale.

The majority of the RRA reports for public health events in a single country were rated as high (55%) or very high (20%) at the national level. For these RRA reports, the regional and global risks were generally from low to moderate (Table 4). However, for regional and multi-regional RRA reports, at the lowest geographical level, more than 50% were assessed as having a high or very high regional risk and, similarly, 76% of global RRA reports were high or very high at the global level (Table 4).

TABLE 4 Risk level by type (global, multi-regional, regional or single country) for Rapid Risk Assessment reports published, 2019–2023, in absolute number and percentage

Country/Regional/Global	National risk level	Regional risk level	Global risk level
Single country (N = 218)			
» Very high	44 (20%)	17 (8%)	4 (2%)
» High	120 (55%)	34 (16%)	3 (1%)
» Moderate	39 (18%)	109 (50%)	2 (1%)
» Low	16 (7%)	59 (27%)	210 (96%)
Regional and multi-regional (N = 18)*			
» Very high	—	0 (0%)	0 (0%)
» High	—	10 (56%)	0 (0%)
» Moderate	—	7 (39%)	3 (23%)
» Low	—	1 (6%)	10 (77%)
Global (N = 25)*			
» Very high	—	—	13 (52%)
» High	—	—	6 (24%)
» Moderate	—	—	6 (24%)
» Low	—	—	0 (0%)

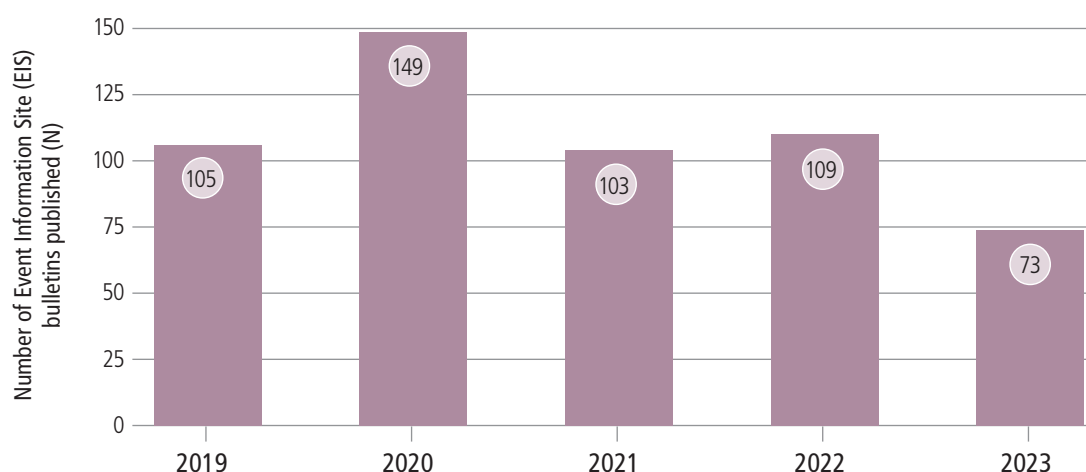
* One global RRA report was not included as it did not assign an overall global risk level and one regional RRA report only assigned a regional, but not an overall global risk level.

2.4 Global information dissemination for acute public health events

Event Information Site postings and announcements

Between 2019 and 2023, there were 539 EIS bulletins published, ranging from 103 to 149 per year. The number of EIS bulletins published per year has been largely constant, except for in 2020 (Fig. 13). Moreover, in the same five-year time period, 487 EIS announcements were published, of which 52 announcements were in 2023 alone.

Most EIS bulletins published during the past five years were about acute public health events in the Western Pacific Region (26%, 136/539), which was closely followed by bulletins for events in the African Region (24%, 129/539). The number of bulletins related to events in the Region of the Americas (17%, 90/539) and the Eastern Mediterranean Region (16%, 86/539) were highly comparable, while fewer bulletins were published in the European Region and the South-East Asia Region, 13% (71/539) and 4% (24/539), respectively.

FIG. 13 Event Information Site bulletins published by year, 2019–2023

The number of EIS bulletins posted annually varied by WHO region; in particular, there was a substantial change for some regions during the first two years of the COVID-19 pandemic (2020–2021) (Table 5).

TABLE 5 Event Information Site bulletins published, 2019–2023, by year and WHO region

WHO Region	2019 N = 105	2020 N = 149	2021 N = 103	2022 N = 109	2023 N = 109	2019–2023 N = 539
AFR	22 (21%)	27 (18%)	26 (25%)	25 (23%)	29 (40%)	129 (24%)
AMR	15 (14%)	40 (27%)	11 (11%)	14 (13%)	10 (14%)	90 (17%)
EMR	34 (32%)	24 (16%)	8 (8%)	16 (15%)	4 (5%)	86 (16%)
EUR	8 (7.6%)	31 (21%)	9 (8%)	13 (12%)	10 (14%)	71 (13%)
SEAR	4 (3.8%)	5 (3%)	4 (4%)	5 (4.6%)	6 (8%)	24 (4%)
WPR	22 (21%)	22 (15%)	45 (44%)	36 (33%)	14 (19%)	139 (26%)
TOTAL	105 (94.4%)	149 (100%)	103 (100%)	109 (100%)	73 (100%)	539 (100%)

WHO regions: AFR: WHO African Region; AMR: WHO Region of the Americas; EMR: WHO Eastern Mediterranean Region; EUR: WHO European Region; SEAR: WHO South-East Asia Region; WPR: WHO Western Pacific Region

Nearly all published EIS bulletins during the five-year time period, except for two, pertained to infectious hazards, of which there were 49 different disease or conditions. The five most common infectious diseases for which an EIS bulletin was published were influenza due to identified avian or animal influenza virus (29%, 156/539), COVID-19 (17%, 90/539), polio (8%, 44/539), Middle-East respiratory syndrome (MERS) (7%, 37/539), dengue fever (5%, 25/539), and cholera (5%, 25/539). More than two-thirds (70%, 377/539) of all EIS bulletins published in the five-year period related to one of these top five infectious diseases (Table 6).

Diseases for which EIS bulletins were published differed by WHO region. For EIS bulletins related to the African Region, yellow fever and cholera were the main diseases. On the other hand, for bulletins in the Region of the Americas and the European Region, it was COVID-19. In the South-East Asia Region, the main diseases were COVID-19 and dengue fever, while for the Western Pacific Region, human cases of influenza due to identified avian or animal influenza virus dominated the EIS bulletins. Finally, MERS was the predominant topic for EIS bulletins in the Eastern Mediterranean Region (Fig. A8–A13).

TABLE 6 Event Information Site bulletins published, by disease, condition or hazard, 2019–2023

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute gastroenteritis	–	–	1	–	1	2
2	Acute haemorrhagic fever syndrome	3	–	2	–	–	5
3	Acute hepatitis E	–	1	4	–	–	5
4	Acute hepatitis of unknown aetiology	–	–	–	1	–	1
5	Anthrax	–	–	–	–	1	1
6	Antibiotic-resistant bacterial infection	2	–	–	–	–	2
7	Arenaviral haemorrhagic fever	–	1	–	–	–	1
8	Argentine haemorrhagic fever	–	1	–	–	–	1
9	Chikungunya	1	1	–	–	–	2
10	Cholera	3	1	4	13	4	25
11	COVID-19	–	90	–	–	–	90
12	Crimean–Congo haemorrhagic fever	–	–	–	1	–	1
13	Dengue fever	10	3	2	7	3	25
14	Diphtheria	–	–	–	–	3	3
15	Dracunculiasis	–	1	–	–	–	1
16	Ebola virus disease	1	4	5	4	–	14
17	Entovirus infection	–	–	–	–	3	3
18	Flood	–	1	–	–	–	1
19	Gonococcal infection	1	–	–	1	–	2
20	Guillain-Barré syndrome	–	–	–	–	1	1
21	Hantavirus pulmonary syndrome	1	–	–	2	–	3
22	Histoplasmosis	1	–	–	–	–	1
23	HIV infection, AIDS	1	–	–	–	–	1
24	Influenza, avian/animal virus	12	16	59	46	23	156

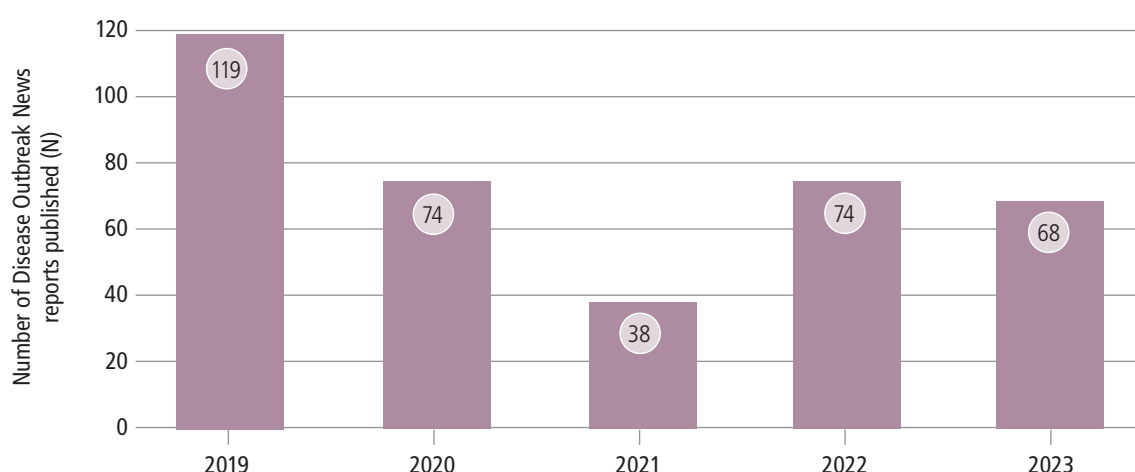
TABLE 6 Event Information Site bulletins published, by disease, condition or hazard, 2019–2023 (*continued*)

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
25	Influenza, human virus	1	–	–	–	2	3
26	Japanese encephalitis	–	–	–	1	–	1
27	Lassa fever	2	2	–	4	1	9
28	Legionellosis	–	–	–	1	1	2
29	Leptospirosis	–	–	–	1	–	1
30	Listeriosis	1	–	–	–	–	1
31	Malaria	1	–	–	1	–	2
32	Marburg virus disease	–	–	2	2	6	10
33	Mayaro virus disease	–	1	–	–	–	1
34	Measles	8	3	–	2	7	20
35	Meningitis	1	–	1	1	3	6
36	Middle East respiratory syndrome	19	6	5	4	3	37
37	Mpox	1	1	4	2	1	9
38	Nipah virus infection	–	–	1	–	2	3
39	Non-infectious unknown aetiology	1	–	–	–	–	1
40	Oropouche virus disease	–	1	–	–	–	1
41	Orthopoxvirus	1	–	–	–	–	1
42	Plague	1	1	2	–	–	4
43	Poliomyelitis	20	3	4	11	6	44
44	Rift Valley fever	2	1	1	1	–	5
45	Rubella	1	–	–	–	–	1
46	Sudan virus disease	–	–	–	1	1	2
47	Western equine encephalitis	–	–	–	–	1	1
48	Yellow fever	8	10	4	2	–	24
49	Zika virus disease	1	–	2	–	–	3

Disease Outbreak News reports

Between 2019 and 2023, 373 DON reports were published, which ranged from 38 to 119 per year. The number of DON reports published annually fluctuated over the past five years, with fewer DON reports during the second year of the COVID-19 pandemic, in 2021. In addition, a very high number of DON reports had been published in 2019 due to regular DON reports for the 2018–2020 Ebola virus disease outbreak. In subsequent years, situation reports were often produced for certain large-scale multi-country public health events, rather than regular DON reports, including for COVID-19, mpox and cholera (Fig. 14).

FIG. 14 Disease Outbreak News reports published, by year, 2019–2023



The majority of DON reports published during the past five years were related to acute public health events in the African Region (50%, 187/373), followed by reports for events in the Eastern Mediterranean Region (16%, 60/373). A comparable percentage of DON reports were published for both the European Region (10%, 37/373) and the Region of the Americas (9%, 34/373). However, fewer DON reports were published for the South-East Asia Region (5%, 18/373) and the Western Pacific Region (5%, 20/373). In addition, during this time period, 4% (15/373) of DON reports were for global acute public health events while 1% (2/373) were for multi-regional events, which affected two WHO regions simultaneously.

During the five-year period, there was a marked variation in the DON reports published annually by WHO region. The number of published DON reports decreased during the first two years of the COVID-19 pandemic (2020–2021) for all WHO regions, except for the South-East Asia Region. In 2023, the number of DON reports was less or similar when compared to the 2020–2021 COVID-19 period with two exceptions: in the Region of the Americas and the South-East Asia Region, the number of DON reports in 2023 was higher than in 2020 or 2021 (Table 7).

TABLE 7 Disease Outbreak News reports published, by year and WHO region, 2019–2023

WHO Region	2019 N = 119	2020 N = 74	2021 N = 38	2022 N = 74	2023 N = 68	2019–2023 N = 373
AFR	66 (55%)	46 (62%)	17 (45%)	28 (38%)	30 (44%)	187 (50.1%)
AMR	7 (5.9%)	7 (9.5%)	4 (11%)	4 (5.4%)	12 (18%)	34 (9.1%)
EMR	28 (24%)	9 (12%)	7 (18%)	12 (16%)	4 (5.9%)	60 (16.1%)
EUR	8 (6.7%)	4 (5.4%)	4 (11%)	11 (15%)	10 (15%)	37 (9.9%)
SEAR	2 (1.7%)	1 (1.4%)	4 (11%)	5 (6.8%)	6 (8.8%)	18 (4.8%)
WPR	7 (5.9%)	6 (8.1%)	1 (2.6%)	2 (2.7%)	4 (5.9%)	20 (5.5%)
Multi-regional*	0 (0%)	0 (0%)	0 (0%)	2 (2.7%)	0 (0%)	2 (0.5%)
Global	1 (0.8%)	1 (1.4%)	1 (2.6%)	10 (14%)	2 (2.9%)	15 (4%)

*Two multi-regional DON reports were both related to AMR and EUR

WHO regions: AFR: WHO African Region; AMR: WHO Region of the Americas; EMR: WHO Eastern Mediterranean Region; EUR: WHO European Region; SEAR: WHO South-East Asia Region; WPR: WHO Western Pacific Region

All DON reports published during the five-year time period were for infectious hazards. There were nearly 50 different infectious disease agents or hazards for which a DON report was published. The five most common diseases were Ebola virus disease (25%, 92/373), MERS (10%, 37/373), poliomyelitis (8%, 28/373), yellow fever (6%, 24/373) and dengue fever (6%, 21/373). These five diseases resulted in half (54%, 202/373) of all DON reports produced in this time period (Table 8).

In addition, for five diseases (dengue fever, MERS, mpox, poliomyelitis and yellow fever) at least one DON report was produced each year between 2019 and 2023 (Fig. A14). Similarly, there were six other diseases (cholera, Ebola virus disease, influenza due to animal/avian or human influenza viruses, measles and Rift Valley fever) which had at least one DON report for four of the five years. Other than these diseases, there was year-to-year variation, with 36 other infectious hazards occurring that necessitated a DON report.

The main disease for which DON reports were published differed by WHO region or geographic reach. In the African Region, the main disease was Ebola virus disease while in the Region of the Americas and the European Region, it was avian human influenza. In the Eastern Mediterranean Region, the main disease was MERS, in the Western Pacific Region it was COVID-19, while in the South-East Asia Region, dengue fever was the main reason for which DON reports were published (Tables A17–A23). In addition, global DON reports were first published in 2019 for measles, which continued in subsequent years and in 2023, included reports on cholera and dengue.

TABLE 8 Disease Outbreak News reports published, by disease, condition or hazard, 2019–2023

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute gastroenteritis	–	–	–	–	1	1
2	Acute hepatitis E	–	–	–	–	1	1
3	Acute hepatitis of unknown aetiology	–	–	–	5	–	5
4	Acute respiratory syndrome	–	–	–	–	1	1
5	Anthrax	–	–	–	–	1	1
6	Antibiotic-resistant bacterial infection	1	–	–	–	–	1
7	Botulism	–	–	–	–	3	3
8	Chikungunya	1	1	–	–	–	2
9	Cholera	1	–	2	10	4	17
10	COVID-19	–	10	–	–	–	10
11	Crimean–Congo haemorrhagic fever	–	–	–	1	–	1
12	Dengue fever	6	3	1	6	5	21
13	Diphtheria	–	–	–	–	3	3
14	Dracunculiasis	–	1	–	–	–	1
15	Ebola virus disease	53	29	6	4	–	92
16	Enterovirus infection	–	–	–	–	3	3
17	Gonococcal infection	1	–	–	–	–	1
18	Guillain-Barré syndrome	–	–	–	–	1	1
19	Hantavirus pulmonary syndrome	2	–	–	–	–	2
20	Hepatitis E	–	1	1	1	–	3
21	HIV infection, AIDS	1	–	–	–	–	1
22	Influenza, avian/animal virus	–	1	2	1	5	9
23	Influenza, human virus	–	1	3	4	7	15
24	Japanese encephalitis	–	–	–	1	–	1
25	Lassa fever	2	1	–	4	1	8
26	Legionellosis	–	–	–	1	1	2
27	Leptospirosis	–	–	–	1	–	1
28	Listeriosis	1	–	–	–	–	1
29	Malaria	–	–	–	1	–	1
30	Marburg virus disease	–	–	2	2	7	11
31	Mayaro fever	–	1	–	–	–	1
32	Measles	8	4	–	2	6	20
33	Meningitis	–	–	1	–	4	5
34	Middle East respiratory syndrome	18	7	5	4	3	37

TABLE 8 Disease Outbreak News reports published by disease, condition or hazard, 2019–2023
(continued)

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
35	Mpox	1	1	4	8	1	15
36	Nipah virus infection	–	–	1	–	2	3
37	Oropouche virus disease	–	1	–	–	–	1
38	Plague	–	1	1	–	–	2
39	Poliomyelitis	13	1	3	6	5	28
40	Rift Valley fever	2	1	1	1	–	5
41	<i>Salmonella typhimurium</i> infection	–	–	–	1	–	1
42	Shigellosis	–	–	–	1	–	1
43	Streptococcus infection	–	–	–	1	–	1
44	Sudan virus disease	–	–	–	5	1	6
45	Western equine encephalitis	–	–	–	–	1	1
46	Yellow fever	7	9	4	3	1	24
47	Zika virus disease	1	–	1	–	–	2

2.5 Public health intelligence in countries requiring international humanitarian assistance

Public health intelligence is as key in countries with humanitarian needs, as it is in other emergency and non-emergency settings. The conditions created by humanitarian crises often increase the epidemic risk, in a context of disruptions to public health interventions, such as vaccination or vector control, population displacement, decreased access to safe water and sanitation, and barriers to accessing healthcare, to name a few. Moreover, large epidemics in humanitarian settings often exacerbate existing health needs by diverting health care and public health resources to the epidemic response. As such, a strong early warning, alert, detection and response (EWAR) function is needed in humanitarian settings to ensure epidemics are detected, and responded to, early.

In 2023, among the 31 countries with a humanitarian response plan (HRP), 27 (87%) were affected by newly reported outbreaks, followed by 10 countries (32%) affected by new natural disasters, three (10%) by societal hazards and three (10%) by product hazards (i.e. counterfeit medicine) (Table 9).

TABLE 9 Countries with a humanitarian response plan by type of hazard, 2023

TYPE OF HAZARD			
INFECTIOUS	DISASTER	SOCIETAL	PRODUCT
Afghanistan	Afghanistan	Cameroon	Cameroon
Bangladesh	Bangladesh	Myanmar	Colombia
Burkina Faso	Democratic Republic of the Congo	Syrian Arab Republic	Honduras
Burundi	Ethiopia		
Central African Republic	Libya		
Chad	Madagascar		
Colombia	Mozambique		
Democratic Republic of the Congo	Myanmar		
El Salvador	Sudan		
Ethiopia	Syrian Arab Republic		
Guatemala			
Honduras			
Libya			
Madagascar			
Mali			
Mongolia			
Mozambique			
Niger			
Nigeria			
Somalia			
South Sudan			
Sudan			
Syrian Arab Republic			
Ukraine			
Venezuela (Bolivarian Republic of)			
Zambia			
Zimbabwe			

In 2023, a total of 85 new events were recorded in the WHO's Event Management System (EMS) across all the 31 countries with HRPs, with the highest number of new events recorded in Colombia (18 new events in EMS in 2023). Of note, 137 events were ongoing across the 31 countries with HRP in 2023.

Communication on events in humanitarian settings is often included in the respective country's situation reports or health cluster bulletins; however, communications under IHR did still occur. Overall, 20 Rapid Risk Assessment reports on acute public health events were undertaken, 36 EIS bulletins for IHR National Focal Points were posted and 16 Disease Outbreak News reports were disseminated across all the 31 countries with HRPs in 2023.

TABLE 10 Key public health intelligence products (EMS events, RRAs, EIS postings, DONs) by country with a humanitarian response plan, 2023

COUNTRY	EVENT (number)		PUBLICATION (number)		
	Ongoing/Acute	New/Recorded in EMS	EIS	RRA	DON
Afghanistan	8	6	0	0	0
Bangladesh	1	3	2	4	2
Burkina Faso	5	2	0	1	0
Burundi	5	2	1	0	1
Cameroon	5	2	0	0	0
Central African Republic	7	1	0	0	0
Chad	6	1	1	1	1
Colombia	0	18	0	0	0
Democratic Republic of the Congo	9	2	2	1	2
El Salvador	0	2	0	0	0
Ethiopia	10	4	1	1	1
Guatemala	0	1	0	0	0
Haiti	0	1	0	0	0
Honduras	0	0	16	0	0
Libya	5	2	0	0	0
Madagascar	5	2	1	0	0
Mali	5	1	0	1	0
Mongolia	0	3	2	0	0
Mozambique	5	2	1	0	1
Myanmar	2	2	0	0	0
Niger	8	1	1	1	1
Nigeria	11	3	4	4	4
Somalia	8	1	0	0	0
South Sudan	7	2	2	0	2
Sudan	8	3	0	3	0
Syrian Arab Republic	1	5	0	0	0
Ukraine	2	1	0	0	0
Venezuela	0	6	0	0	0
Yemen	6	0	0	0	0
Zambia	3	2	1	3	1
Zimbabwe	5	4	1	0	0

3. Emerging trends for key diseases

The year 2023 marked the end of mpox and COVID-19 as PHEICs, coinciding with the lifting of the restrictions implemented by many countries, resulting in a return to population mobility and mixing. This was accompanied by the re-emergence of pathogens in populations with immunity gaps and increased susceptibility, resulting in public health events that affected primarily the paediatric population with increased severity being reported to WHO. This included the increase in cases of severe neonatal sepsis associated with enterovirus (Echovirus-11, E-11), the increased incidence of scarlet fever and invasive Group A *Streptococcus* infection, and the upsurge of respiratory illnesses among children due to circulating known pathogens such as influenza, *Mycoplasma pneumoniae* and respiratory syncytial virus (RSV).

The year 2023 also marked the global resurgence of cholera, characterized by the number, size and concurrence of multiple outbreaks, the spread to areas free of cholera for decades and alarmingly high mortality rates, as well as the global upsurge in dengue fever cases, characterized by the scale and simultaneous occurrence of multiple outbreaks, spreading into areas previously unaffected by dengue fever.

Key disease trend in the WHO African Region

In 2023, a total of 112 substantiated acute public health events were verified and reported across 37 Member States in the WHO African Region. Senegal and Uganda reported the highest number of events (8 each), followed by Ethiopia, Guinea and Mauritania (6 each). The most prevalent acute public health event reported was dengue fever (13 cases), followed by cholera (12 cases) and floods (9 cases).

In 2023, cholera emerged as the major outbreak in the African Region with a total of 193 330 cholera cases reported across 17 countries in the Region. Climate change has played a significant role in exacerbating the cholera outbreak. The surge in precipitation levels resulted in devastating floods and landslides, within communities heightening the vulnerability of countries that had not yet reported cases, and increasing the likelihood of outbreaks.

The African Region is strongly affected by arboviruses, including yellow fever, dengue fever, chikungunya, O'nyong nyong, Rift Valley fever and Zika virus disease. In 2023, 171 991 dengue fever cases were reported in countries in the Region with 753 deaths. Evidence of dengue fever circulation was detected in local populations and/or among travellers returning from more than 30 African countries. The most affected country in the Region in 2023 was Burkina Faso, experiencing a significant increase in dengue fever cases compared with the same periods in 2021 and 2022.

Key disease trend in the WHO Region of the Americas

During 2023, public health faced several challenges across the Region due to the occurrence of multiple events that affected the general population. These challenges included a notable increase in dengue fever surpassing previous epidemic years, increases in chikungunya in some countries, the emergence of autochthonous cases of malaria in areas where cases were not previously registered, the persistence of the cholera outbreak in Hispaniola, and outbreaks of avian influenza A(H5N1) affecting birds and mammals, with isolated cases identified in humans. Additionally, in the Southern Cone countries, there were reports of Western equine encephalitis in horses, with subsequent occurrence of isolated cases in humans. Yellow fever outbreaks were also identified, alongside an outbreak of health-care-associated meningitis and an outbreak of Guillain-Barré syndrome. A case of vaccine-derived polio virus type 2 (PVDV2) was also detected. Furthermore, there was an increase in cases of acute respiratory illness related to the circulation of influenza and other respiratory viruses, especially in countries of the Southern Cone.

Key disease trend in the WHO Eastern Mediterranean Region

In 2023, the Eastern Mediterranean Region faced many challenges, including cholera outbreaks in several countries, including Afghanistan, Somalia, Sudan and Yemen. These outbreaks were driven by floods, heavy rains or droughts in many of these countries. This environmental phenomenon has also resulted in a dramatic increase in the trend of vector-borne diseases, especially dengue fever, in many countries. Other vaccine-preventable disease outbreaks were also reported across some of the Region's countries, including measles and diphtheria, underscoring the need to fill the gap in vaccination coverage in those countries.

Key disease trend in the WHO European Region

In the WHO European Region in 2023, a significant portion of the public health events were related to the humanitarian crises due to conflicts (e.g. the war in Ukraine and related-refugee crisis, and Israel and the occupied Palestinian territories conflict), as well as natural disasters (e.g. the Türkiye–Syria earthquake). There have been reports of emerging and re-emerging diseases among refugees and displaced populations, in particular vaccine preventable diseases such as diphtheria, pertussis and measles, likely due to delays or refusals of routine immunizations. There were also reports of increased detections of highly pathogenic avian influenza H5 among wild birds, poultry and mammals (including fur farmed animals and cats), as well as among humans. However, human cases are still considered sporadic and mild, and some were associated with heightened active surveillance and case finding among exposed populations. Vector-borne diseases were also reported in an increased frequency including an expanded geographical spread of autochthonous cases in the southern part of the Region (e.g. France, Italy and Spain), along with West Nile infections during the transmission season.

Key disease trend in the WHO South-East Asia Region

Dengue fever continues to pose a high burden in the Region. In 2023, Bangladesh experienced a substantial increase in dengue fever cases and deaths, 4.8 times and 9.3 times the average of recent years, respectively. Thailand also recorded a surge in dengue fever cases and deaths, 2.3 times and 2.5 times the average number of the recent five years, respectively.

At least 496 342 COVID-19 cases and 4 997 related deaths have been recorded in 2023 in the Region, although both are a substantial reduction (over 90%) compared to 2022. Nevertheless, countries recorded case surges from around April to July 2023, coinciding with the introduction of the XBB.1.16 sub-variant. Influenza case surges were recorded in Bangladesh (peaking in August) and Thailand (peaking in October) in 2023.

Nipah virus disease outbreaks occurred in Bangladesh (12 confirmed and 2 probable cases including 10 deaths) and in Kerala State, India, (six cases including two deaths) in 2023. Thailand and Indonesia recorded a surge of mpox cases, with 673 cases and 72 cases reported in 2023, respectively.

Vaccine-preventable diseases also caused outbreaks, including events due to vaccine-derived polio virus type 2 (Indonesia), measles (Indonesia, Nepal, Sri Lanka) and diphtheria (Indonesia). Events of acute watery diarrhoea, associated with cholera, were reported in both Bangladesh and India.

The Region is also prone to natural hazards — in particular, cyclone Mocha (Myanmar) and an earthquake in western Nepal that caused major damage in 2023.

Key disease trend in the WHO Western Pacific Region

In 2023, human infections with avian influenza A including H5N1, H3N8, H5N6 and H9N2 were reported from China. Severe illness and death associated with the avian influenza virus were also reported. The Philippines experienced a cholera outbreak, with a total of 3756 cholera cases and 19 deaths from 1 January 2023 to 31 December 2023. Dengue fever remains endemic in several countries in the Region and in 2023, over 500 000 dengue cases and 750 deaths were reported from eight countries: Australia, Cambodia, China, Lao People's Democratic Republic, Malaysia, Philippines, Singapore and Viet Nam.

4. Conclusion

This report highlights the significance of public health intelligence activities for the global detection, verification, risk assessment and information dissemination of acute public health events under the IHR (2005) framework, in collaboration with States Parties. It also highlights the multitude of health hazards identified worldwide and across different WHO regions, the comprehensive risk assessments conducted by WHO, and meticulous dissemination of information to both States Parties and the general public to avert potential PHEIC.

In 2023, nearly 1500 signals of potential public health events of international concern were detected globally, for which 422 requests for verification were sent. In one aspect, this represented a significant decline in the number of signals identified, but it is a reflection of a revised signal definition and triaging criteria implemented in 2023 to ensure better synchronization with operational procedures. Conversely, the number of verification requests was considerably higher compared to the preceding five years.

Tracking efforts for public health incidents rely heavily on timeliness, especially when it comes to detecting, notifying and responding to acute events. From 2019 to 2023, the average time to detect events was 4 days, the average time to notify WHO was 2 days, and the average time to respond was 0 days. Although there is some improvement in responding promptly to signal verifications over the past two years, there is still room for improvement in notifying WHO promptly.

In 2023, a total of 365 new events were recorded in EMS, including both verified signals and events directly reported under the IHR (2005) by NFPs. All events recorded in EMS were assessed based on public health risk and a further 52 in depth RRA reports were produced and disseminated. In addition, in 2023, 73 EIS bulletins and 52 EIS announcements were distributed, which preceded the dissemination of 68 DON reports. Furthermore, additional information products were developed and disseminated by each of the WHO regions for timely information sharing.

In the past five years, events reported from the African Region and the Region of the Americas have accounted for approximately 20–30% of all events, while the events in each of the other WHO regions generally have ranged from 5% to 20%. Different factors contribute to the variations in events documented by each WHO Region, such as the number of countries, territories and areas within each region. When evaluating the percentage of countries, territories and areas that reported acute public health incidents annually over the past five years, the majority of WHO regions had a minimum of two-thirds of their respective countries, territories and areas reporting at least one acute public health event.

Infectious diseases were the predominant cause of events in the past five years; in 2023, infectious diseases accounted for 80% (289) of all events globally that were reported to WHO. Of note, events caused by disasters have been increasingly recorded, particularly in

the African, Eastern Mediterranean, South-East Asia and Western Pacific Regions, whereas in the Region of the Americas and the European Region, events due to product-related hazards have been more frequently recorded in recent years.

In 2023, most acute public health events of an infectious nature were largely due to dengue fever, followed by measles, cholera, influenza due to identified avian or animal influenza virus and poliomyelitis. While infectious diseases account for the majority of health threats worldwide and in many WHO regions, non-infectious disease threats are also prevalent and can have a significant impact on public health.

The diverse range of hazards on a global scale and across various WHO regions underscores the significance of an all-encompassing approach to hazards, as outlined in the IHR (2005). Given the complex nature of health threats, interdisciplinary teams with diverse expertise are crucial to analyse and evaluate these threats. A crucial aspect in this context is the utilization of the Joint FAO–WHO–WOAH Global Early Warning System (GLEWS+) by PHI teams to monitor health threats and emerging risks at the human-animal-ecosystem interface.

The number of events reported to WHO directly by the NFP or national government agency in 2023 decreased compared to the previous year. Despite this, in the past five years, an increasing trend in the NFP and national government agency as the initial source of events has been observed in the Region of the Americas, Eastern Mediterranean Region, European Region and Western Pacific Region. Protracted events such as COVID-19 and mpox may have affected the capacity of NFP or national government agencies to detect, respond and report to new and emerging threats.

In order to enhance the initial reporting process, both regional and country offices consistently reach out to Member States to verify signals and enhance the initial reporting process emphasizing the significance of maintaining cooperative relationships. This further underlines the significance of WHO's collection of public health information to identify and verify signals in WHO regions where the initial information provided by NFPs or national governments is limited.

Information dissemination to alert Member States to public health threats represents a crucial part of the reporting phase of the public health intelligence cycle. In the past five years, 262 RRA reports, 539 EIS bulletins, 487 EIS announcements and 373 DON reports were disseminated. The number of EIS bulletins decreased markedly in 2023, reaching the lowest number in the last five years ($n = 73$). The number of RRAs and DONs reports published in 2023 remains aligned with the previous five-year period.

A limitation of this report is represented by the fact that it does not reflect the true number of acute public health events that occur continuously across the globe. WHO and national governments detect a multitude of events daily, not all of which are required to be reported to WHO. A further limitation pertains to the fact that the data on signals are only available at WHO HQ level for more detailed analysis, moving forward it is hoped that signal data from all levels will be available to provide a more comprehensive picture on WHO's collective early detection processes.

Furthermore, the report solely concentrates on reported events within a specific year, disregarding the possibility of events spanning multiple years. Consequently, there is an

underestimation of the impact of long-lasting acute public health events that countries, territories or areas might face. Another constraint arises from the varying data entry protocols and data management practices across WHO Regions when it comes to verification requests and event recording in EMS. These discrepancies may contribute to the differences observed in certain findings between WHO regions. Nevertheless, there are continuous endeavours to enhance the standardization of operating procedures throughout WHO regions. Despite these disparities, the utilization of EMS and other WHO platforms continues to be crucial for swift information exchange and effective data handling.

Regarding information products, this report primarily concentrates on RRA reports, EIS bulletins and announcements, and DON reports. Nevertheless, WHO's PHI teams are engaged in numerous other analyses and information products. These encompass regional information products like risk assessments, bulletins, situation reports and interactive dashboards. Consequently, the conclusions drawn from RRA reports, EIS bulletins and announcements and DON reports merely encompass a small portion of the extensive range of information products that PHI teams oversee or participate in as a means of alerting the wider public health community to events of concern.

In conclusion, this report emphasizes the significance of PHI activities in the identification, verification, evaluation of risks and dissemination of information regarding acute public health events worldwide. The primary objective of PHI is to detect health threats at an early stage and provide guidance for decision-making and response efforts. The COVID-19 pandemic, along with other recent large-scale outbreaks like cholera and dengue fever, underscores the crucial role of public health intelligence in the global health framework.

In an era marked by increasing health risks, the resurgence of illnesses, conflicts and the growing influence of climate change, it is imperative and crucial to provide ongoing support and improvement to PHI. This is vital for effectively tackling and alleviating forthcoming health crises.

PHI is of utmost importance in the global health framework and in the World Health Organization's efforts to address health crises worldwide. In light of the escalating health risks, an even stronger and resilient global public health intelligence infrastructure will be indispensable.

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Annexes

Annex 1

States Parties by WHO region

Annex 2

Supplementary figures

Annex 3

Supplementary tables

Annex 4

Regional information dissemination for acute public health events

NOTE

For all the figures/tables in the Annexes, the legends for the WHO regions are:

AFR: WHO African Region

AMR: WHO Region of the Americas

EMR: WHO Eastern Mediterranean Region

EUR: WHO European Region

SEAR: WHO South-East Asia Region

WPR: WHO Western Pacific Region

Annex 1

States Parties and areas by WHO Region

WHO African Region

The WHO African Region consists of the following 47 States Parties: Algeria, Angola, Benin, Botswana, Burkina Faso, Burundi, Cabo Verde, Cameroon, Central African Republic, Chad, Comoros, Congo, Côte d'Ivoire, Democratic Republic of the Congo, Equatorial Guinea, Eritrea, Eswatini, Ethiopia, Gabon, Gambia, Ghana, Guinea, Guinea-Bissau, Kenya, Lesotho, Liberia, Madagascar, Malawi, Mali, Mauritania, Mauritius, Mozambique, Namibia, Niger, Nigeria, Rwanda, Sao Tome and Principe, Senegal, Seychelles, Sierra Leone, South Africa, South Sudan, Togo, Uganda, United Republic of Tanzania, Zambia, Zimbabwe.

WHO Region of the Americas

The WHO Region of the Americas consists of the following 35 States Parties: Antigua and Barbuda, Argentina, Bahamas, Barbados, Belize, Bolivia (Plurinational State of), Brazil, Canada, Chile, Colombia, Costa Rica, Cuba, Dominica, Dominican Republic, Ecuador, El Salvador, Grenada, Guatemala, Guyana, Haiti, Honduras, Jamaica, Mexico, Nicaragua, Panama, Paraguay, Peru, Puerto Rico, Saint Kitts and Nevis, Saint Lucia, Saint Vincent and the Grenadines, Suriname, Trinidad and Tobago, United States of America, Uruguay, Venezuela (Bolivarian Republic of).

WHO Eastern Mediterranean Region

The WHO Eastern Mediterranean Region consists of the following 21 States Parties and the occupied Palestinian territory, including east Jerusalem: Afghanistan, Bahrain, Djibouti, Egypt, Iran (Islamic Republic of), Iraq, Jordan, Kuwait, Lebanon, Libya, Morocco, Oman, Pakistan, Qatar, Saudi Arabia, Somalia, Sudan, Syrian Arab Republic, Tunisia, United Arab Emirates, Yemen.

WHO European Region

The WHO European Region consists of the following 55 States Parties: Albania, Andorra, Armenia, Austria, Azerbaijan, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Georgia, Germany, Greece, Holy See, Hungary, Iceland, Ireland, Israel, Italy, Kazakhstan, Kyrgyzstan, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Monaco, Montenegro, Netherlands (Kingdom of the), North Macedonia, Norway, Poland, Portugal, Republic of Moldova, Romania, Russian Federation, San Marino, Serbia, Slovakia, Slovenia, Spain, Sweden, Switzerland, Tajikistan, Türkiye, Turkmenistan, Ukraine, United Kingdom of Great Britain and Northern Ireland, Uzbekistan.

WHO South-East Asia Region

The WHO South-East Asia Region consists of the following 11 States Parties: Bangladesh, Bhutan, Democratic People's Republic of Korea, India, Indonesia, Maldives, Myanmar, Nepal, Sri Lanka, Thailand, Timor-Leste.

WHO Western Pacific Region

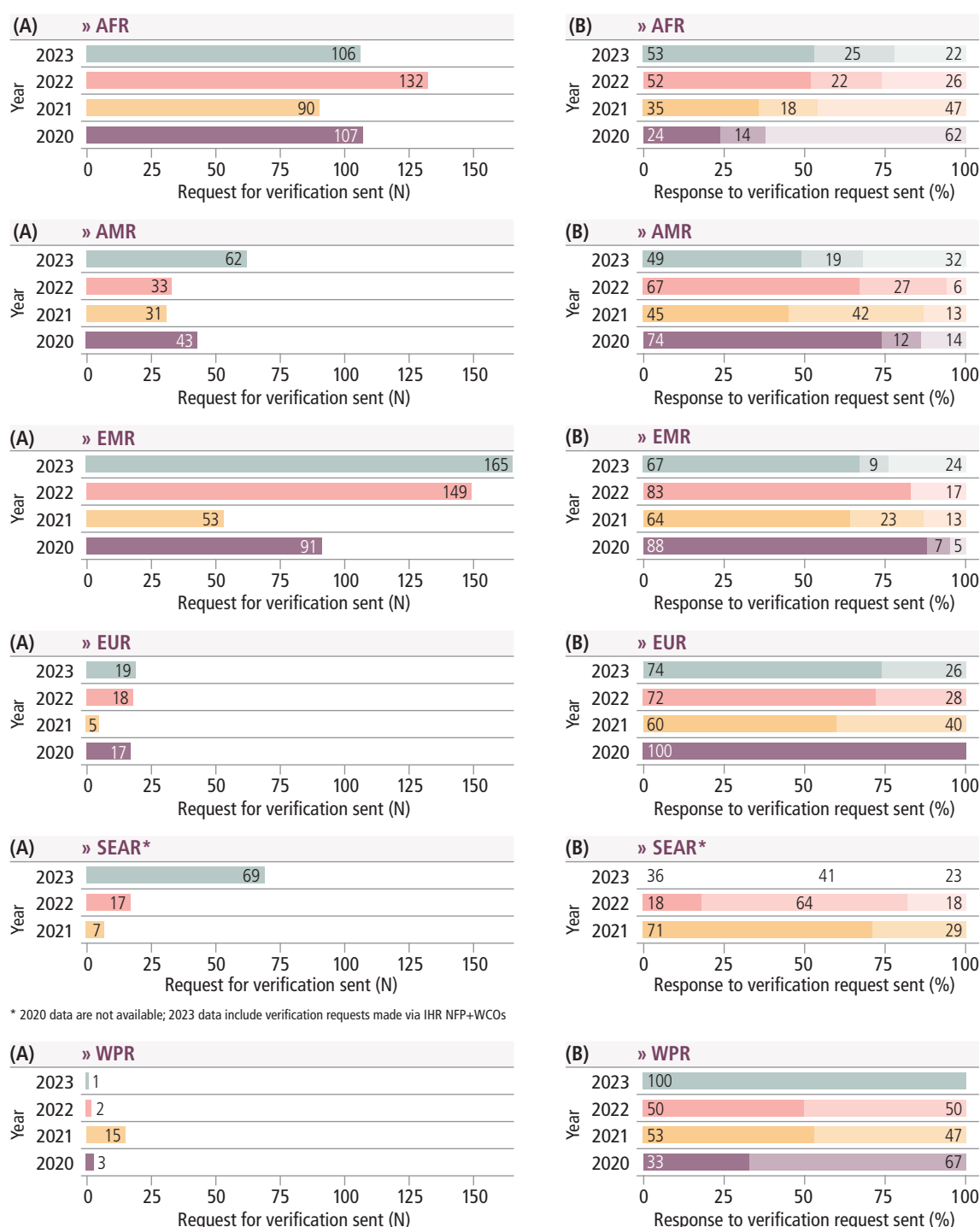
The WHO Western Pacific Region consists of the following 37 States Parties and areas: American Samoa (USA), Australia, Brunei Darussalam, Cambodia, China, Cook Islands, Fiji, French Polynesia (France), Guam (USA), Hong Kong SAR (China), Japan, Kiribati, Lao People's Democratic Republic, Macao SAR (China), Malaysia, Marshall Islands, Micronesia (Federated States of), Mongolia, Nauru, New Caledonia (France), New Zealand, Niue, Northern Mariana Islands [Commonwealth of the, (USA)], Palau, Papua New Guinea, Philippines, Pitcairn Island (UK), Republic of Korea, Samoa, Singapore, Solomon Islands, Tokelau (New Zealand), Tonga, Tuvalu, Vanuatu, Viet Nam, Wallis and Futuna (France).

Annex 2: Supplementary figures

WHO regions

AFR: WHO African Region, AMR: WHO Region of the Americas, EMR: WHO Eastern Mediterranean Region, EUR: WHO European Region, SEAR: WHO South-East Asia Region, WPR: WHO Western Pacific Region

FIG. A1 (A) Requests for verification sent (N) and (B) responses to verification requests (%) by WHO Region, 2020–2023



* 2020 data are not available; 2023 data include verification requests made via IHR NFP+WCOs

Response time: ≤ 48 h > 48 h No response

FIG. A2 Initial source of information for events reported globally by year and WHO region, 2019–2023



NFP: National International Health Regulations Focal Point

Source of information: ■ WHO ■ NFP and national government

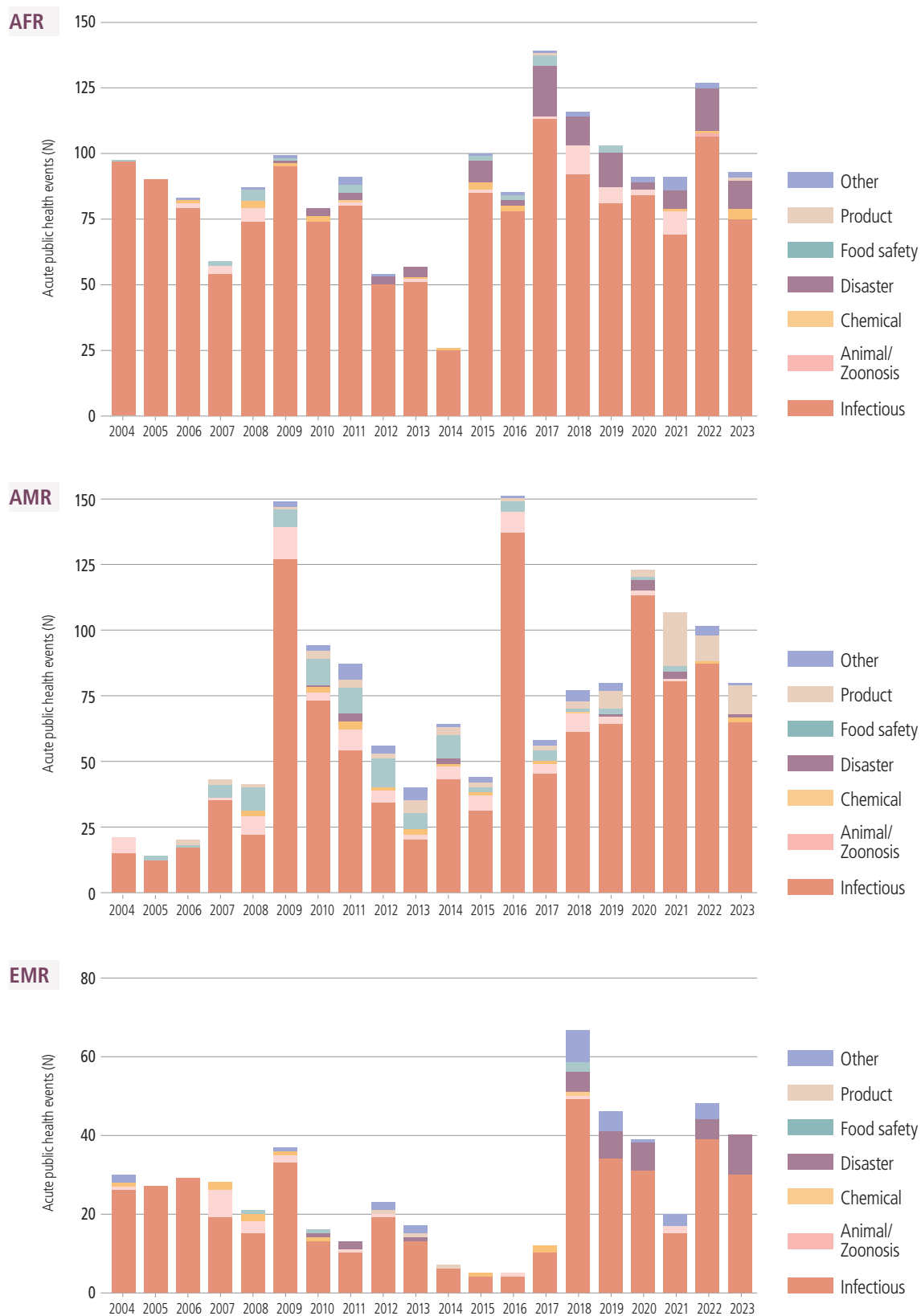
FIG. A3 Acute public health events, in absolute number, by hazard type and WHO region, 2004–2023

FIG. A3 Acute public health events, in absolute number, by hazard type and WHO region, 2004–2023 (*continued*)

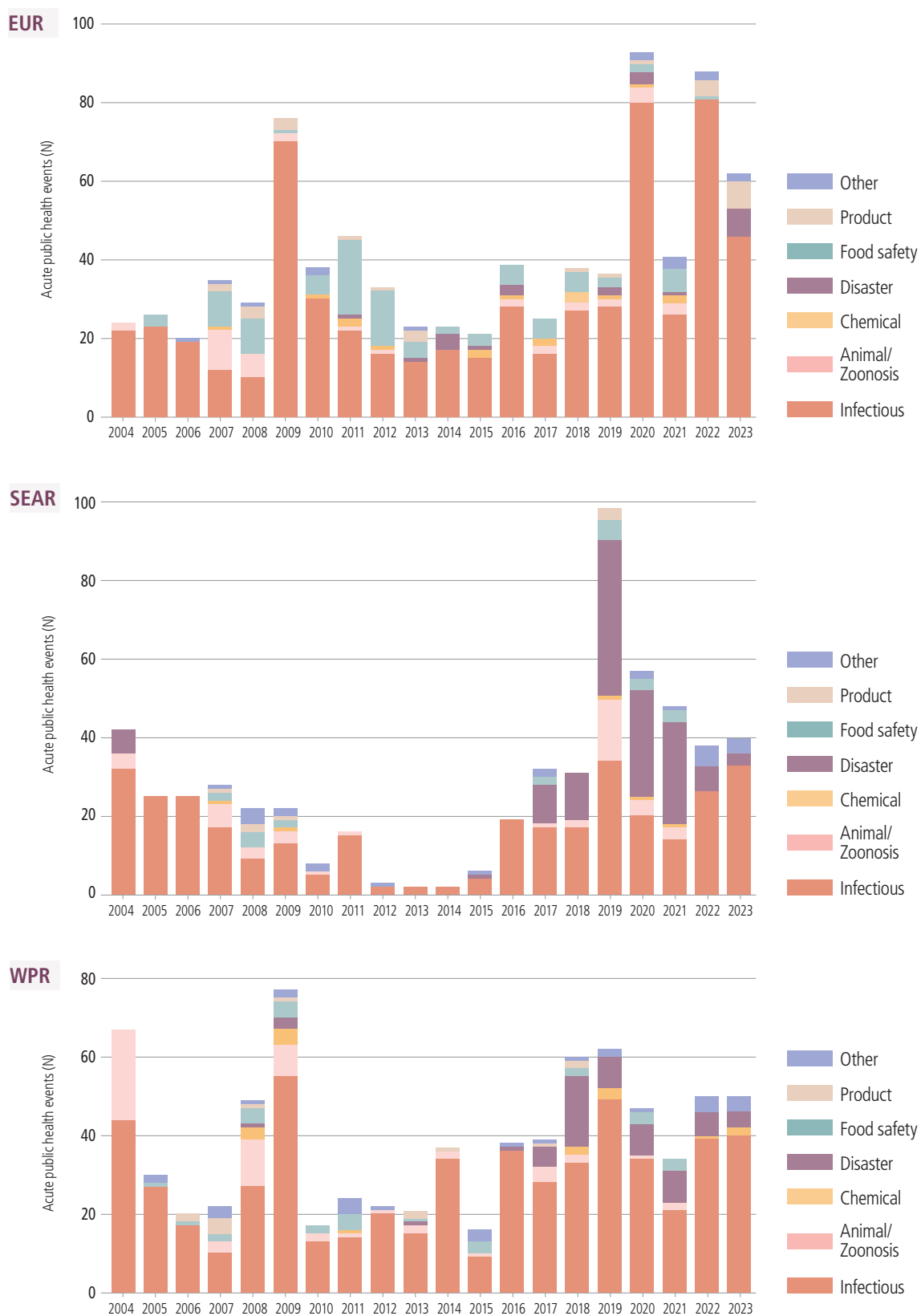


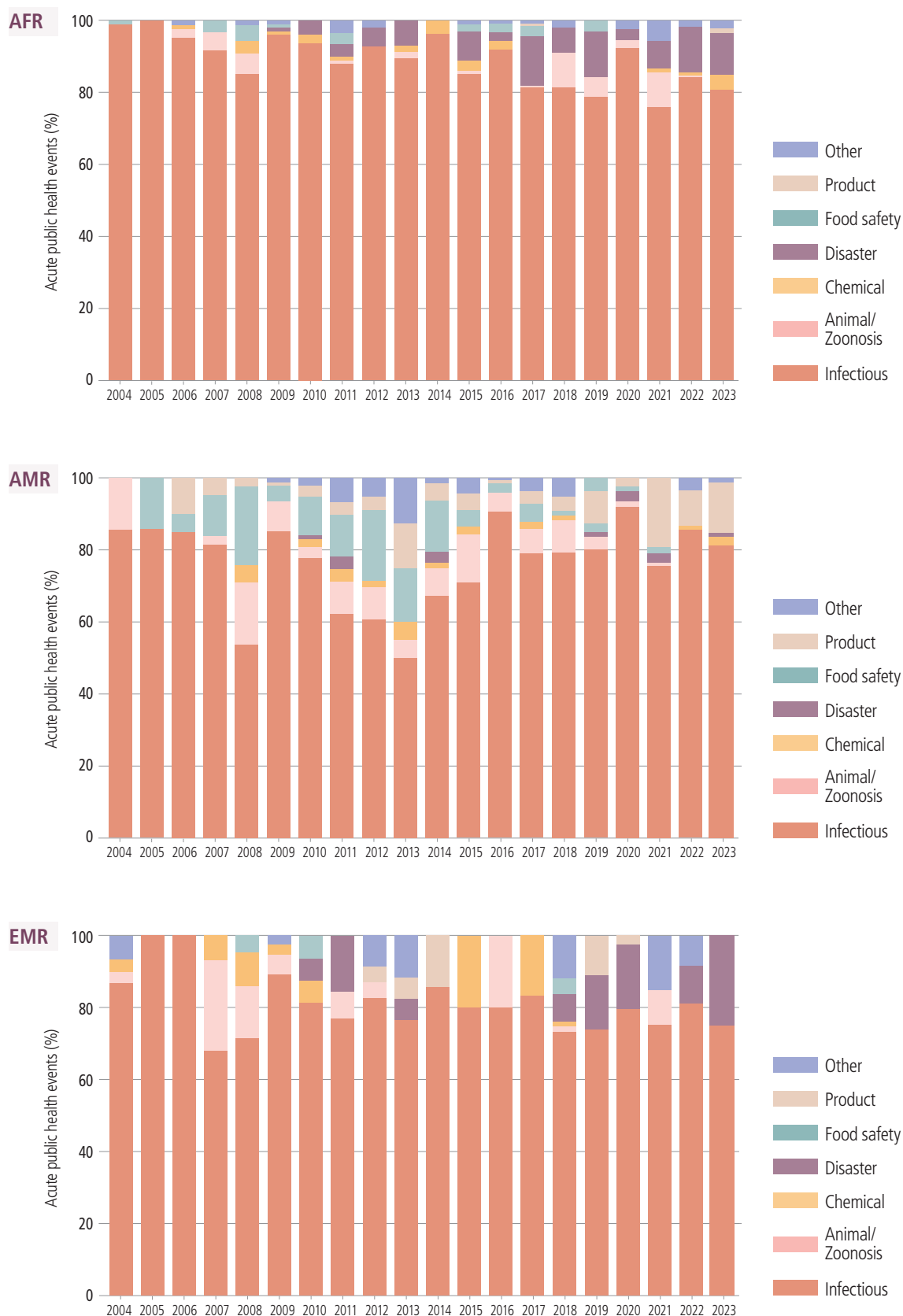
FIG. A4 Acute public health events, in percentage, by hazard type and WHO region, 2004–2023

FIG. A4 Acute public health events, in percentage, by hazard type and WHO region, 2004–2023
(continued)

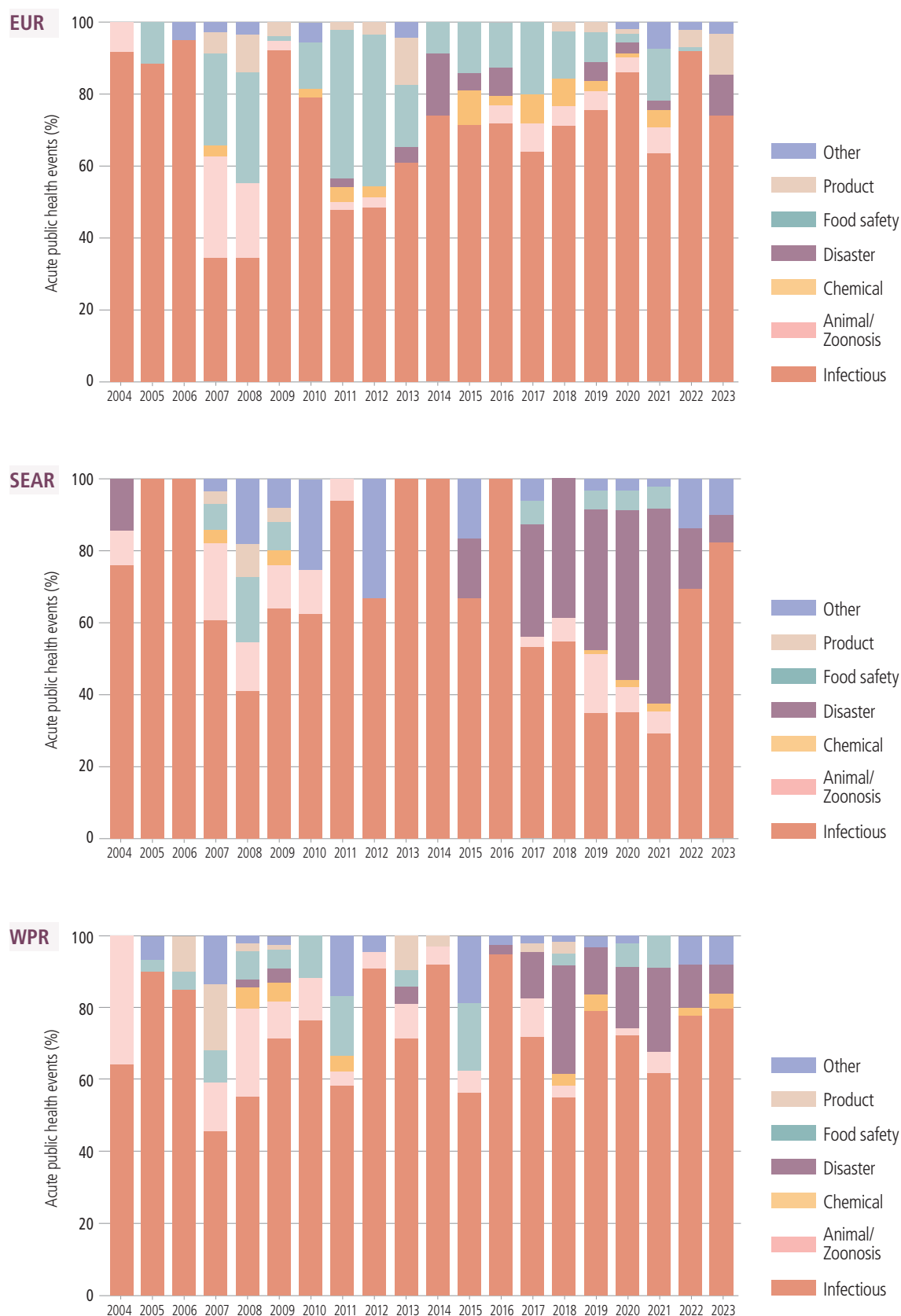


FIG. A5 Timeliness (in days) of events published in the Event Information Site for detection, notification and response, by WHO region, 2019–2023

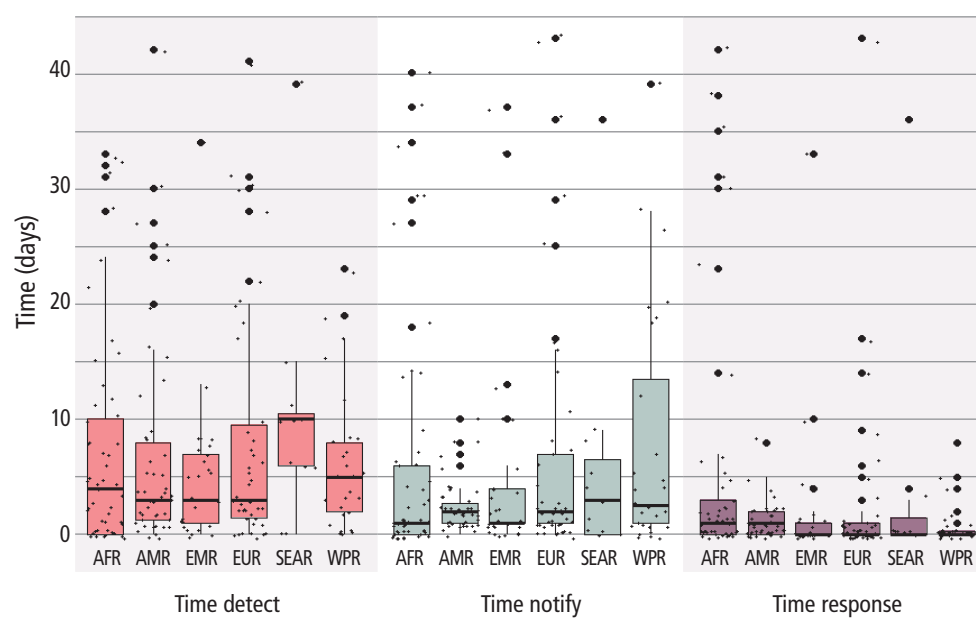
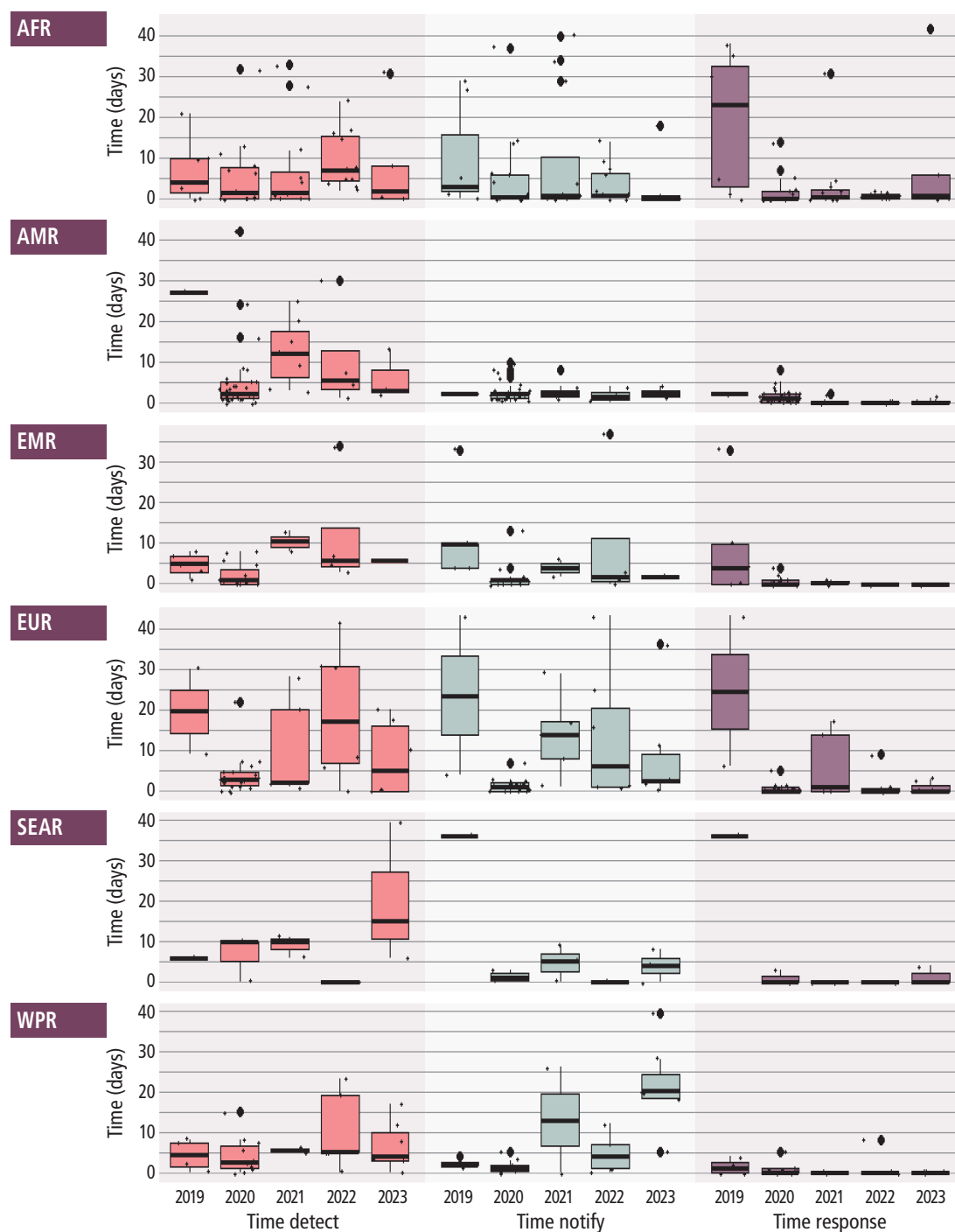


FIG. A6 Timeliness (in days) of events published in the Event Information Site for detection, notification and response, by WHO region, 2019–2023



Annex 3: Supplementary tables

WHO regions

AFR: WHO African Region, **AMR:** WHO Region of the Americas, **EMR:** WHO Eastern Mediterranean Region, **EUR:** WHO European Region, **SEAR:** WHO South-East Asia Region, **WPR:** WHO Western Pacific Region

TABLE A1 Acute public health events by hazard type and WHO region, 2019–2023*

Hazard category	AFR N = 505	AMR N = 488	EMR N = 193	EUR N = 321	SEAR N = 278	WPR N = 243
Animal/Zoonosis	15 (3%)	6 (1.2%)	2 (1%)	9 (2.8%)	23 (8.3%)	3 (1.2%)
Chemical	6 (1.2%)	3 (0.6%)	0 (0%)	4 (1.2%)	3 (1.1%)	6 (2.5%)
Disaster	51 (10%)	9 (1.8%)	30 (16%)	13 (4%)	100 (36%)	34 (14%)
Food safety	3 (0.6%)	5 (1%)	0 (0%)	12 (3.7%)	11 (4%)	6 (2.5%)
Infectious	417 (83%)	406 (83%)	149 (77%)	260 (81%)	126 (45%)	183 (75%)
Other	12 (2.4%)	5 (1.0%)	12 (6.2%)	9 (2.8%)	15 (5.4%)	7 (2.9%)
Product	1 (0.2%)	54 (11%)	0 (0%)	14 (4.4%)	0 (0%)	4 (1.6%)

* Since 2022, food safety and animal/zoonosis hazards have been reclassified to other corresponding hazards, explaining the absence of these hazard types in 2023.

TABLE A2 Acute public health events by disease type, 2019–2023

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	—	—	—	1	—	1
2	Acute kidney injury	—	—	—	1	—	1
3	Acute watery diarrhoea	1	—	—	—	—	1
4	Anthrax	—	—	—	—	2	2
5	Antibiotic-resistant bacterial infection	2	—	—	1	—	3
6	Argentine haemorrhagic fever	—	1	—	—	—	1
7	Chikungunya	2	1	—	—	1	4
8	Cholera	6	3	8	19	6	42
9	COVID-19	—	11	5	5	3	24
10	Crimean–Congo haemorrhagic fever	—	1	—	1	1	3
11	Dengue fever	6	—	2	4	7	19

TABLE A2 Acute public health events by disease type, 2019–2023 (*continued*)

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
12	Diphtheria	—	1	—	—	4	5
13	Ebola virus disease	11	6	4	2	—	23
14	Environmental contamination	—	1	—	—	—	1
15	Gastroenteritis	—	—	—	—	1	1
16	Hantavirus pulmonary syndrome	1	—	—	—	—	1
17	Hepatitis E	—	2	2	—	—	4
18	HIV infection	1	—	1	—	—	2
19	Infectious acute febrile syndrome	—	—	1	—	—	1
20	Influenza, avian/animal virus	—	2	1	—	3	6
21	Japanese encephalitis	—	—	—	1	—	1
22	Lassa fever	3	1	—	3	1	8
23	Leptospirosis	—	—	—	1	—	1
24	Malaria	3	—	2	—	1	6
25	Marburg virus disease	—	—	1	1	3	5
26	Measles	9	1	—	9	6	25
27	Meningococcal disease/Meningitis	1	—	1	4	3	9
28	Middle East respiratory syndrome	—	—	—	—	1	1
29	Mpox	—	1	—	4	2	7
30	Nipah virus infection	2	—	1	—	2	5
31	Plague	—	1	2	—	—	3
32	Poliomyelitis	6	—	2	2	2	12
33	Rabies	—	—	—	—	1	1
34	Rift Valley fever	1	1	—	1	—	3
35	Risk of biological hazard	—	—	—	—	2	2
36	<i>Salmonella typhimurium</i> infection	—	—	—	1	—	1
37	Smallpox vaccine	1	—	—	—	—	1
38	Sudan virus disease	—	—	—	2	—	2
39	Viral haemorrhagic fever	—	1	—	—	—	1
40	Whooping cough (pertussis)	1	—	—	—	—	1
41	Yellow fever	5	6	5	2	—	18
42	Zika virus disease	1	—	2	—	—	3

TABLE A3 Acute public health events for the most frequently reported diseases, conditions or hazards of an infectious nature, by WHO region, for each year of the past five years and aggregated across the five-year period 2019–2023

AFR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Anthrax	5	–	–	7	6	19
2	Chikungunya	3	2	–	–	2	7
3	Cholera	16	5	13	15	12	61
4	COVID-19	–	48	–	–	–	48
5	Crimean–Congo haemorrhagic fever	5	4	2	7	5	23
6	Dengue fever	8	1	6	5	8	28
7	Diphtheria	–	–	–	–	5	5
8	Dracunculiasis	–	1	–	–	–	1
9	Ebola virus disease	–	1	2	–	–	3
10	Lassa fever	6	–	5	7	–	21
11	Malaria	–	2	2	–	–	4
12	Measles	12	3	2	16	5	38
13	Meningitis	3	–	2	6	–	14
14	Mpox	–	–	–	9	–	9
15	Poliomyelitis	9	–	10	5	5	30
16	Rift Valley fever	3	–	–	–	3	3
17	Yellow fever	–	9	7	7	4	29
AMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	–	–	–	10	–	10
2	Anthrax	–	–	1	–	–	1
3	Antibiotic resistant agents	3	1	–	–	–	4
4	Arenaviral haemorrhagic fevers, other	2	–	2	–	–	4
5	Bacterial diseases, other specified	2	4	3	–	2	12
6	Chikungunya	2	–	–	–	3	5
7	Cholera	–	–	–	3	–	3
8	COVID-19	–	57	38	–	–	95
9	Dengue fever	8	9	–	2	12	31
10	Diphtheria	–	3	2	–	–	5
11	Gastroenteritis	4	–	–	–	–	4
12	Guillain-Barré syndrome	–	–	–	–	2	2

TABLE A3 (Continued)

AMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
13	Hantavirus pulmonary syndrome	–	–	–	2	–	2
14	Influenza, avian/animal virus	–	3	8	6	–	17
15	Influenza, human virus	3	–	–	–	2	5
16	<i>Klebsiella pneumoniae</i>	–	–	–	–	2	2
17	Malaria	–	2	–	–	4	6
18	Measles	8	2	3	2	3	18
19	Meningitis	–	–	–	3	–	3
20	Mpox	–	–	–	31	–	31
21	Multisystem inflammatory syndrome in children (MIS-C)	–	18	8	–	–	26
22	Rabies	–	–	5	4	3	12
23	Rubella	3	–	–	–	–	3
24	Shigellosis	–	–	–	3	–	3
25	Yellow fever	3	5	3	–	4	15
EMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute diarrhoeal disease outbreak	–	–	2	–	–	2
2	Acute hepatitis of unknown aetiology	–	–	–	1	–	1
3	Chikungunya	2	1	–	–	–	3
4	Cholera	3	–	1	5	2	11
5	COVID-19	–	22	–	–	–	22
6	Crimean–Congo haemorrhagic fever	1	–	–	1	2	4
7	Dengue fever	5	1	3	5	4	18
8	Diphtheria	2	–	–	2	1	5
9	Gastroenteritis	–	1	–	–	–	1
10	Health impact from armed conflict	–	–	–	–	1	1
11	Hepatitis A	–	–	–	1	–	1
12	Hepatitis E	–	–	1	–	–	1
13	HIV infection, AIDS	2	–	–	–	–	2
14	Legionellosis	–	–	–	–	3	3
15	Malaria	2	1	–	3	–	5
16	Measles	4	–	5	3	5	17
17	MERS–CoV	6	1	2	2	2	13
18	Mpox	–	–	–	10	2	12
19	Poliomyelitis	2	4	1	–	2	9

TABLE A3 (Continued)

EUR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Anthrax	1	–	1	–	–	2
2	Antibiotic resistant agents	1	–	–	–	–	1
3	Bacterial diseases, other specified	–	1	–	–	–	1
4	Botulism	1	1	2	–	2	6
5	<i>Burkholderia</i> spp.	–	–	–	–	1	1
6	Cholera	–	1	–	8	1	10
7	COVID-19	–	62	–	–	–	62
8	Crimean–Congo haemorrhagic fever	–	–	1	4	–	5
9	Dengue	4	2	1	2	2	11
10	Detection of New Delhi metallo– β -lactamase (NDM) producing organisms	1	–	–	–	–	1
11	Diphtheria	1	–	–	9	–	10
12	<i>Escherichia coli</i> infection	1	–	1	–	–	2
13	Influenza, avian/animal virus	–	1	3	4	10	18
14	Legionellosis	–	1	–	–	2	3
15	Malaria	–	3	2	–	–	5
16	Measles	3	–	–	–	3	6
17	Meningitis	–	–	–	3	–	3
18	Meningococcal disease	–	–	–	–	3	3
19	Mpox	–	–	–	23	–	23
20	Poliomyelitis	2	2	7	5	7	23
21	Salmonella infection	–	–	3	–	3	6
22	Shigellosis	–	–	–	3	–	3
23	West Nile fever	2	2	2	3	–	9
SEAR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Adenovirus infection	–	–	–	–	1	1
2	Anthrax	–	–	–	–	1	1
3	Chikungunya	1	–	–	–	1	2
4	Cholera	3	–	5	8	3	19
5	COVID-19	–	11	–	1	–	12
6	Crimean–Congo haemorrhagic fever	3	–	–	–	–	3
7	Dengue fever	7	2	3	5	4	21
8	Diphtheria	–	2	–	–	2	4
9	Gastroenteritis	–	1	–	–	–	1
10	Hand, foot and mouth disease	–	–	–	1	–	1
11	Hepatitis A	1	–	–	–	–	1

TABLE A3 (Continued)

SEAR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
12	Influenza, avian/animal virus	1	–	2	2	–	5
13	Influenza, human virus	4	–	1	–	–	5
14	Measles	4	2	–	2	4	12
15	Mpox	–	–	–	4	2	6
16	Mumps	–	1	–	–	–	1
17	Nipah virus infection	–	–	–	–	2	2
18	Poliomyelitis	2	–	–	1	–	3
19	Viral encephalitis	3	1	1	–	–	5
20	Zika virus disease	–	–	2	1	2	5
WPR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute diarrhoeal disease outbreak	–	–	–	1	–	1
2	Acute respiratory infection	–	–	–	–	2	2
3	Adenovirus infection	–	–	–	1	–	1
4	Anthrax	–	–	–	–	2	2
5	Botulism	–	–	–	–	2	2
6	Chikungunya	1	1	1	–	–	3
7	Cholera	1	–	2	5	–	8
8	COVID-19	–	26	6	4	3	39
9	Dengue	10	1	2	4	3	20
10	Diphtheria	–	1	–	–	3	4
11	Food poisoning	–	–	–	1	–	1
12	Gastroenteritis	–	–	–	2	–	2
13	Hepatitis A	–	–	1	–	–	1
14	Influenza, avian/animal virus	–	2	3	5	6	16
15	Influenza, human virus	4	–	–	–	–	4
16	Japanese encephalitis	–	–	–	1	–	1
17	Leptospirosis	2	–	1	–	–	3
18	Measles	14	1	1	–	–	16
19	Meningococcal disease	2	–	–	–	–	2
20	Mpox	–	–	–	13	4	17
21	Plague	2	2	–	–	4	8
22	Poliomyelitis	4	–	1	–	–	5
23	Rotavirus	–	–	2	–	–	2
24	Whooping cough (pertussis)	2	–	–	–	–	2
25	Zika virus disease	–	–	–	–	3	3

RAPID RISK ASSESSEMENT (RRA) REPORTS

Tables from A5 to A10: RRA reports published by disease, condition or hazard and year, 2019–2023, or across the five-year time period for acute public health events in the six WHO regions.

TABLE A4 Rapid Risk Assessment reports published by disease, condition or hazard, for acute public health events across all WHO regions, 2019–2023

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019*	2020	2021	2022	2023	
1	Cholera	—	—	—	1	3	4
2	COVID-19	—	3	5	4	3	15
3	Dengue fever	—	—	—	—	1	1
4	Mpox	—	—	—	4	2	6

* There were no global RRA reports published in this year. However, it is very important to note that RRAs are a subset of all risk assessments that WHO conducts.

TABLE A5 Rapid Risk Assessment reports published by disease, condition or hazard, for acute public health events in the WHO African Region, 2019–2023

AFR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute gastroenteritis	—	—	1	—	—	1
2	Acute kidney injury	—	—	—	1	—	1
3	Anthrax	—	—	—	—	2	2
4	Chikungunya	2	1	—	—	—	3
5	Cholera	4	3	6	8	3	24
6	COVID-19	—	1	—	—	—	1
7	Crimean–Congo haemorrhagic fever	—	1	—	—	—	1
8	Dengue fever	—	—	1	1	3	5
9	Diphtheria	—	—	—	—	3	3
10	Ebola virus disease	11	6	4	2	—	23
12	Gastroenteritis	—	—	—	—	1	1
11	Hepatitis E	—	2	1	—	—	3
13	Lassa fever	2	1	—	3	1	7

TABLE A5 Rapid Risk Assessment reports published by disease, condition or hazard, for acute public health events in the WHO African Region, 2019–2023 (*continued*)

AFR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
14	Leptospirosis	–	–	–	1	–	1
15	Malaria	1	–	2	–	–	3
16	Marburg virus disease	–	–	1	1	3	5
17	Measles	3	2	–	4	2	10
18	Meningococcal disease/Meningitis	–	–	1	4	3	8
19	Mpox	–	1	–	–	–	1
20	Plague	–	1	2	–	–	3
21	Poliomyelitis	3	–	–	–	–	3
22	Rift Valley fever	–	1	–	1	–	2
23	Sudan virus disease	–	–	–	2	–	2
24	Viral haemorrhagic fever	–	1	–	–	–	1
25	Yellow fever	3	6	4	2	–	15

TABLE A6 Rapid Risk Assessment reports published by disease, condition or hazard, for acute public health events in the WHO Region of the Americas, 2019–2023

AMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	–	–	–	1	–	1
2	Antibiotic-resistant bacteria	1	–	–	–	–	1
3	Chikungunya	–	–	–	–	1	1
4	Cholera	–	–	–	2	–	2
5	Dengue fever	–	–	–	–	1	1
6	Hantavirus pulmonary syndrome	1	–	–	–	–	1
7	Influenza, avian/animal virus	–	–	–	–	1	1
8	Malaria	1	–	–	–	–	1
9	Poliomyelitis	–	–	–	1	1	2
10	<i>Salmonella typhimurium</i>	–	–	–	1	–	1
11	Smallpox vaccine	1	–	–	–	–	1
12	Whooping cough (pertussis)	1	–	–	–	–	1
13	Yellow fever	2	–	1	–	–	3

TABLE A7 Rapid Risk Assessment reports published by disease, condition or hazard, for acute public health events in the WHO Eastern Mediterranean Region, 2019–2023

EMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020*	2021	2022	2023	
1	Acute gastroenteritis	–	–	1	–	–	1
2	Cholera	2	–	2	7	–	11
3	Crimean–Congo haemorrhagic fever	–	–	–	1	–	1
4	Dengue fever	3	–	1	1	1	6
5	Hepatitis E	–	–	1	–	–	1
6	HIV infection	1	–	1	–	–	2
7	Influenza, avian/animal virus	–	–	1	–	–	1
8	Malaria	1	–	–	–	1	2
9	Measles	4	–	–	5	1	10
10	Middle East respiratory syndrome	–	–	–	–	1	1
11	Rift Valley Fever	1	–	–	–	–	1
12	Risk of biological hazard	–	–	–	–	2	2

* There were no global RRA reports published in this year for this Region. However, it is important to note that RRAs are a subset of all risk assessments that WHO conducts.

TABLE A8 Rapid Risk Assessment reports published by disease, condition or hazard, for acute public health events in the WHO European Region, 2019–2023

EUR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	–	–	–	1	–	1
2	Antibiotic-resistant bacterial infection	1	–	–	1	–	2
3	Argentine haemorrhagic fever	–	1	–	–	–	1
4	Influenza, avian/animal virus	–	–	–	–	1	1
5	COVID-19	–	2	–	–	–	2
6	Crimean–Congo haemorrhagic fever	–	–	–	–	1	1
7	Dengue fever	1	–	–	–	–	1
8	Lassa fever	1	–	–	–	–	1
9	Poliomyelitis	–	–	2	–	–	2
10	<i>Salmonella typhimurium</i>	–	–	–	1	–	1
11	Zika virus disease	1	–	–	–	–	1

TABLE A9 Rapid Risk Assessment reports published by disease, condition or hazard, for acute public health events in the WHO South-East Asia Region, 2019–2023

SEAR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute watery diarrhoea	1	–	–	–	–	1
2	Cholera	–	–	–	1	–	1
3	COVID-19	–	–	1	1	–	2
4	Dengue fever	2	–	–	2	1	5
5	Diphtheria	–	1	–	–	1	2
6	Environmental contamination	–	1	–	–	–	1
7	Measles	–	–	–	–	3	3
8	Nipah virus infection	2	–	1	–	2	5
9	Poliomyelitis	2	–	–	1	1	4
10	Rabies	–	–	–	–	1	1
11	Zika virus disease	–	–	2	–	–	2

TABLE A10 Rapid Risk Assessment reports published by disease, condition or hazard, for acute public health events in the WHO Western Pacific Region, 2019–2023

WPR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021*	2022	2023	
1	COVID-19	–	5	–	–	–	5
2	Influenza, avian/animal virus	–	2	–	–	1	3
3	Japanese encephalitis	–	–	–	1	–	1
4	Measles	2	–	–	–	–	2
5	Meningococcal disease/Meningitis	1	–	–	–	–	1
6	Poliomyelitis	1	–	–	–	–	1

* There were no global RRA reports published in this year for this Region. However, it is important to note that RRAs are a subset of all risk assessments that WHO conducts.

EVENT INFORMATION SITE (EIS) BULLETINS

Tables from A11 to A16: EIS bulletins published by disease, condition or hazard and year, or across the five-year time period 2019–2023, for acute public health events in the six WHO regions.

TABLE A11 Event Information Site bulletins published by disease, condition or hazard, for acute public health events in the WHO African Region, 2019–2023

AFR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute gastroenteritis	–	–	1	–	1	2
2	Acute haemorrhagic fever syndrome	3	–	2	–	–	5
3	Acute hepatitis E	–	1	3	–	–	4
4	Anthrax	–	–	–	–	1	1
5	Chikungunya	1	1	–	–	–	2
6	Cholera	2	1	4	6	4	17
7	COVID-19	–	3	–	–	–	3
8	Dengue fever	–	–	1	1	1	3
9	Diphtheria	–	–	–	–	3	3
10	Dracunculiasis	–	1	–	–	–	1
11	Ebola virus disease	1	4	5	4	–	14
12	Influenza, avian/animal virus	–	–	1	–	–	1
13	Lassa fever	1	2	–	3	1	7
14	Leptospirosis	–	–	–	1	–	1
15	Marburg virus disease	–	–	2	2	6	10
16	Measles	3	2	–	–	3	8
17	Meningitis	–	–	1	1	3	5
18	Mpox	–	1	–	–	1	2
19	Plague	–	1	2	–	–	3
20	Poliomyelitis	6	–	–	3	4	13
21	Rift Valley fever	–	1	1	1	–	3
22	Sudan virus disease	–	–	–	1	1	2
23	Yellow fever	5	9	3	2	–	19

TABLE A12 Event Information Site bulletins published by disease, condition or hazard, for acute public health events in the WHO Region of the Americas, 2019–2023

AMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Antibiotic-resistant bacterial infection	2	–	–	–	–	2
2	Arenaviral haemorrhagic fever	–	1	–	–	–	1
3	Cholera	–	–	–	2	–	2
4	COVID-19	–	29	–	–	–	29
5	Dengue fever	3	2	–	–	–	5
6	Guillain-Barré syndrome	–	–	–	–	1	1
7	Hantavirus pulmonary syndrome	1	–	–	2	–	3
8	Histoplasmosis	1	–	–	–	–	1
9	Influenza, avian/animal virus	1	4	8	5	3	21
10	Influenza, human virus	–	–	–	–	2	2
11	Legionellosis	–	–	–	1	–	1
12	Malaria	1	–	–	–	–	1
13	Mayaro virus disease	–	1	–	–	–	1
14	Measles	–	1	–	–	2	3
15	Mpox	–	–	2	–	–	2
16	Oropouche virus disease	–	1	–	–	–	1
17	Orthopoxvirus	1	–	–	–	–	1
18	Poliomyelitis	2	–	–	4	1	7
19	Western equine encephalitis	–	–	–	–	1	1
20	Yellow fever	3	1	1	–	–	5

TABLE A13 Event Information Site bulletins published by disease, condition or hazard, for acute public health events in the WHO Eastern Mediterranean Region, 2019–2023

EMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis E	–	–	1	–	–	1
2	Cholera	1	–	–	5	–	6
3	COVID-19	–	16	–	–	–	16
4	Crimean–Congo haemorrhagic fever	–	–	–	1	–	1
5	Dengue fever	5	–	1	2	1	9
6	Flood	–	1	–	–	–	1
7	HIV infection, AIDS	1	–	–	–	–	1

TABLE A13 Event Information Site bulletins published by disease, condition or hazard, for acute public health events in the WHO Eastern Mediterranean Region, 2018–2022 (*continued*)

EMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
8	Influenza, avian/animal virus	1	–	–	–	–	1
9	Malaria	–	–	–	1	–	1
10	Measles	4	–	–	2	–	6
11	Middle East respiratory syndrome	19	6	5	4	3	37
12	Poliomyelitis	2	1	1	1	–	5
13	Rift Valley fever	1	–	–	–	–	1

TABLE A14 Event Information Site bulletins published by disease, condition or hazard, for acute public health events in the WHO European Region, 2019–2023

EUR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	–	–	–	1	–	1
2	Argentine haemorrhagic fever	–	1	–	–	–	1
3	COVID-19	–	26	–	–	–	26
4	Dengue fever	2	1	–	–	–	3
5	Enterovirus infection	–	–	–	–	3	3
6	Gonococcal infection	1	–	–	1	–	2
7	Influenza, avian/animal virus	–	2	5	6	6	19
8	Influenza, human virus	1	–	–	–	–	1
9	Lassa fever	1	–	–	1	–	2
10	Legionellosis	–	–	–	–	1	1
11	Listeriosis	1	–	–	–	–	1
12	Mpox	–	–	2	2	–	4
13	Poliomyelitis	–	1	2	2	–	5
14	Rift Valley fever	1	–	–	–	–	1
15	Zika virus disease	1	–	–	–	–	1

TABLE A15 Event Information Site bulletins published by disease, condition or hazard, for acute public health events in the WHO South-East Asia Region, 2019–2023

SEAR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	COVID-19	–	5	–	–	–	5
2	Dengue fever	–	–	–	4	1	5
3	Influenza, avian/animal virus	2	–	1	–	–	3
4	Measles	–	–	–	–	2	2
5	Nipah virus infection	–	–	1	–	2	3
6	Poliomyelitis	2	–	–	1	1	4
7	Zika virus disease	–	–	2	–	–	2

TABLE A16 Event Information Site bulletins published by disease, condition or hazard, for acute public health events in the WHO Western Pacific Region, 2019–2023

WPR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	COVID-19	–	11	–	–	–	11
2	Influenza, avian/animal virus	8	10	44	35	14	111
3	Japanese encephalitis	–	–	–	1	–	1
4	Measles	1	–	–	–	–	1
5	Meningitis	1	–	–	–	–	1
6	Mpox	1	–	–	–	–	1
7	Non-infectious unknown aetiology	1	–	–	–	–	1
8	Plague	1	–	–	–	–	1
9	Poliomyelitis	8	1	1	–	–	10
10	Rubella	1	–	–	–	–	1

DISEASE OUTBREAK NEWS (DON) REPORTS

Tables from A18 to A23: DON reports published by disease, condition or hazard and year, 2019–2023, or across the five-year time period for acute public health events in the six WHO regions

TABLE A17 Disease Outbreak News reports published by disease, condition or hazard, for acute public health events across all WHO regions, 2019–2023

No.	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	–	–	–	3	–	3
2	Cholera	–	–	–	1	1	2
3	COVID-19	–	1	–	–	–	1
4	Dengue fever	–	–	–	–	1	1
5	Measles	1	–	–	–	–	1
6	Mpox	–	–	–	6	–	6
7	Poliomyelitis	–	–	1	–	–	1

TABLE A18 Disease Outbreak News reports published by disease, condition or hazard, for acute public health events in the WHO African Region, 2019–2023

AFR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute gastroenteritis	–	–	–	–	1	1
2	Acute hepatitis E	–	–	–	–	1	1
3	Anthrax	–	–	–	–	1	1
4	Chikungunya	1	1	–	–	–	2
5	Cholera	–	–	2	4	3	9
6	Dengue fever	–	–	–	1	1	2
7	Diphtheria	–	–	–	–	3	3
8	Dracunculiasis	–	1	–	–	–	1
9	Ebola virus disease	53	29	6	4	–	92
10	Hepatitis E	–	1	1	1	–	3
11	Lassa fever	1	1	–	3	1	6
12	Leptospirosis	–	–	–	1	–	1

TABLE A18 Disease Outbreak News reports published by disease, condition or hazard, for acute public health events in the WHO African Region, 2019–2023 (*continued*)

AFR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
13	Marburg virus disease	–	–	2	2	7	11
14	Measles	1	2	–	–	3	6
15	Meningitis	–	–	1	–	3	4
16	Mpox	–	1	–	–	1	2
17	Plague	–	1	1	–	–	2
18	Poliomyelitis	6	–	–	3	3	12
19	Rift Valley fever	–	1	1	1	–	3
20	Sudan virus disease	–	–	–	5	1	6
21	Yellow fever	4	8	3	3	1	19

TABLE A19 Disease Outbreak News reports published by disease, condition or hazard, for acute public health events in the WHO Region of the Americas, 2019–2023

AMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	–	–	–	1	–	1
2	Antibiotic-resistant bacterial infection	1	–	–	–	–	1
3	Cholera	–	–	–	2	–	2
4	Dengue fever	1	2	–	–	2	5
5	Guillain-Barré Syndrome	–	–	–	–	1	1
6	Hantavirus pulmonary syndrome	2	–	–	–	–	2
7	Influenza, <i>avian/animal</i> virus	–	–	–	–	2	2
8	Influenza, <i>human</i> virus	–	1	2	1	3	7
9	Legionellosis	–	–	–	1	–	1
10	Mayaro fever	–	1	–	–	–	1
11	Measles	–	1	–	–	2	3
12	Meningitis	–	–	–	–	1	1
13	Mpox	–	–	1	–	–	1
14	Oropouche virus disease	–	1	–	–	–	1
15	Poliomyelitis	–	–	–	1	–	1
16	Western equine encephalitis	–	–	–	–	1	1
17	Yellow fever	3	1	1	–	–	5

TABLE A20 Disease Outbreak News reports published by disease, condition or hazard, for acute public health events in the WHO Eastern Mediterranean Region, 2019–2023

EMR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Cholera	1	–	–	3	–	4
2	Crimean–Congo haemorrhagic fever	–	–	–	1	–	1
3	Dengue fever	3	–	1	1	–	5
4	HIV infection, AIDS	1	–	–	–	–	1
5	Malaria	–	–	–	1	–	1
6	Measles	2	1	–	2	–	5
7	Middle East respiratory syndrome	18	7	5	4	3	37
8	Poliomyelitis	2	1	1	–	1	5
9	Rift Valley fever	1	–	–	–	–	1

TABLE A21 Disease Outbreak News reports published by disease, condition or hazard, for acute public health events in the WHO European Region, 2019–2023

EUR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute hepatitis of unknown aetiology	–	–	–	2	–	2
2	Botulism	–	–	–	–	2	2
3	COVID-19	–	3	–	–	–	3
4	Dengue fever	2	1	–	–	–	3
5	Enterovirus infection	–	–	–	–	3	3
6	Gonococcal infection	1	–	–	–	–	1
7	Influenza, avian/animal virus	–	–	–	1	1	2
8	Influenza, human virus	–	–	1	2	3	6
9	Lassa fever	1	–	–	1	–	2
10	Legionellosis	–	–	–	–	1	1
11	Listeriosis	1	–	–	–	–	1
12	Measles	1	–	–	–	–	1
13	Mpox	–	–	2	2	–	4
14	Poliomyelitis	–	–	1	2	–	3
15	Rift Valley fever	1	–	–	–	–	1
16	<i>Salmonella typhimurium</i> infection	–	–	–	1	–	1
17	Shigellosis	–	–	–	1	–	1
18	Streptococcus infection	–	–	–	1	–	1
19	Zika virus disease	1	–	–	–	–	1

TABLE A22 Disease Outbreak News reports published by disease, condition or hazard, for acute public health events in the WHO South-East Asia Region, 2019–2023

SEAR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Botulism	–	–	–	–	1	1
2	COVID-19/	–	1	–	–	–	1
3	Dengue fever	–	–	–	4	1	5
4	Influenza, avian/animal virus	–	–	1	–	–	1
5	Measles	–	–	–	–	1	1
6	Mpox	–	–	1	–	–	1
7	Nipah virus infection	–	–	1	–	2	3
8	Poliomyelitis	2	–	–	1	1	4
9	Zika virus disease	–	–	1	–	–	1

TABLE A23 Disease Outbreak News reports published by disease, condition or hazard, for acute public health events in the WHO Western Pacific Region, 2019–2023

WPR	Disease/Condition/Hazard	Annually					Five-year period 2019–2023
		2019	2020	2021	2022	2023	
1	Acute respiratory syndrome	–	–	–	–	1	1
2	COVID-19	–	5	–	–	–	5
3	Influenza, avian/animal virus	–	1	1	–	2	4
4	Influenza, human virus	–	–	–	1	1	2
5	Japanese encephalitis	–	–	–	1	–	1
6	Measles	3	–	–	–	–	3
7	Mpox	1	–	–	–	–	1
8	Poliomyelitis	3	–	–	–	–	3

Annex 4

Regional information dissemination for acute public health events

There are numerous regional information products that provide essential and timely information on acute public health events within the WHO regions. Annex 4 provides a non-exhaustive list of these by WHO region.

WHO African Region

- » Weekly bulletins on outbreaks and other emergencies
- » Cholera weekly bulletin
- » Weekly regional Diphtheria Bulletin
- » Dengue fever regional report (monthly)

WHO Region of the Americas

- » Epidemiological Alerts and Updates
- » Briefing and technical notes and laboratory guidelines
- » Daily Natural Hazards Monitoring
- » Public health signals and events being monitored in the Region of the Americas Dashboard
- » SARS-CoV-2 situation in the Region of the Americas Dashboard
- » El Niño Southern Oscillation Disease Dashboard
- » Mpox cases in the Region of the Americas Dashboard
- » Cholera resurgence in Hispaniola Dashboard
- » Western equine encephalitis Dashboard

WHO Eastern Mediterranean Region

- » Weekly sitreps for Sudan escalation of violence
- » Weekly sitreps for Humanitarian impact of the current situation in OPT
- » Morocco Earthquake information for briefing notes
- » Daily bulletins and weekly bulletins

- » EMRO WHE “go public” bulletin
- » EMRO health emergencies quarterly bulletin
- » Morocco earthquake sitreps
- » Türkiye–Syria earthquake sitreps

WHO European Region

- » Quarterly Operational Update for EURO (for all graded emergencies)
- » Internal Response Dashboard + OSL (for all graded emergencies)
- » Quarterly Ukraine Bulletin (including refugee-hosting countries)
- » External Situation Report for Ukraine
- » External Situation Report for the refugee crisis in Armenia (weekly, then twice a month, until discontinued)
- » External Situation Report for the Earthquake in Türkiye (weekly, then twice a month, then discontinued)
- » Monthly donor report (done by the WHO Office in Gaziantep) for NW Syria
- » Year-in-Review for COVID-19 in EURO (summer 2023)
- » ECDC Daily and Weekly Communicable Diseases Threat Reports
- » The European Respiratory Virus Surveillance Summary (ERVISS)

WHO South-East Asia Region

- » WHO South-East Asia Region Epidemiological Bulletin
- » Dashboard: COVID-19 Situation in the WHO South-East Asia Region
- » WHO South-East Asia Region Weekly Situation Updates (internal)
- » Public Health Situation Analysis (PHSA) – Myanmar, August 2023
- » Technical Brief: Enhancing readiness for a Nipah virus event in countries not reporting a Nipah virus event
- » Health Emergency Information and Risk Assessment

WHO Western Pacific Region

- » Avian Influenza weekly update
- » Dengue situation updates
- » Seasonal influenza update
- » COVID-19 situation report

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